I.

On the Stability of Equilibrium.

By G. Lejeune Dirichlet.

Crelle, Journal für die reine und angewandte Mathematik, vol. 32, pp. 85–88.

On the Stability of Equilibrium.

[Extract from a paper read in the Royal Academy of Sciences on 22 January 1846.][1]

When attractive or repulsive forces, depending only on distance, act on a system of material points, either directed toward fixed centers, or such that, when they occur mutually between two masses, action and reaction are equal; and when, moreover, the constraint equations connecting the coordinates of the points with one another do not contain the time, then the so-called integral of vis viva always holds. In its full generality it was first set out by D. Bernoulli, and is contained in the equation

$$\Sigma mv^2 = f(x, y, z, x', \ldots) + C.$$

The summation sign extends over all masses of the system; each is denoted by m, its velocity by v, and C denotes the arbitrary constant. The function of the coordinates depends only on the nature of the forces and can always be expressed by a definite number of independent variables $\lambda, \mu, \nu, \ldots$, whereby the equation becomes

$$\Sigma mv^2 = \varphi(\lambda, \mu, \nu, \ldots) + C.$$

The function φ is intimately related to the equilibrium positions of the system, since the condition that equilibrium should take place for a position corresponding to specified values of $\lambda, \mu, \nu, \ldots$ coincides with the vanishing of the total differential of φ for these values; thus, in general, for every equilibrium position the function will have a maximum or a minimum. If a maximum actually occurs, then the equilibrium has the character of stability; that is, if the points are displaced only slightly from such a position and receive small initial velocities, the system will never, in the course of time, move away from that position beyond certain narrow bounds.

The theorem just mentioned is unquestionably among the most important in all mechanics; in particular, the theory of small oscillations, which contains so many interesting physical applications, rests upon it. One must indeed wonder that its truth has not until now been demonstrated with the necessary rigor. Suppose that the equilibrium position, or the maximum of the function φ, corresponds to the values $\lambda = 0$, $\mu = 0$, $\ldots$; this can always be done without loss of generality. The proof given by Lagrange[2] consists in expanding the function in powers of $\lambda, \mu, \nu, \ldots$; since this expansion begins with terms of the second order, he restricts it to those terms, and then from the known criterion for a maximum, according to which the second-order terms can be represented as a sum of negative squares, derives bounds for $\lambda, \mu, \nu, \ldots$ which they cannot exceed. This line of reasoning, which is also not seldom applied to other questions of stability, especially in physical astronomy, is plainly without demonstrative force. It can rightly be doubted whether quantities for which, under the supposition that they always remain small — for only in this supposition lies the permission to neglect the higher terms —, one finds small

[1] In the 1846 volume of the Academy reports the extract is introduced on p. 34 with the words: Mr. Lejeune Dirichlet read *on the conditions for the stability of equilibrium.* For the printing in the Journal, Dirichlet changed the title, as well as the text in a few places, and newly added the entire final paragraph. K.

[2] Mécanique analytique, première partie, section III.

bounds, will after an arbitrary time actually be enclosed within these, or even within narrow bounds at all.

So far as I know, the proof just mentioned has hitherto been reproduced without essential alteration by all writers who have dealt with this subject; and what Poisson[3] in particular added to it, in order to take account of the terms of higher order, rests on the quite untenable assumption that each term of the second order exceeds the aggregate of all higher terms. But even if Lagrange's considerations were completed for the case to which they apply, and in which the maximum is revealed in the terms of second order, the theorem in question would still not thereby be proved in its full extent. As is well known, the existence of a maximum is compatible with the vanishing of the second-order terms; only the lowest order for which not all terms vanish must always be even, and the aggregate of all terms of that order must always remain negative. The complete criteria for this latter condition have not yet been established, even when the fourth order is concerned (Théorie des fonctions, seconde partie, no. 57);[4] they would therefore have to be sought first, which would necessarily introduce great complication into the proof of the mechanical theorem.

Fortunately, the principle of the stability of equilibrium can be established independently of these criteria, and by very simple considerations directly connected with the concept of a maximum.

Keeping the above supposition that the equilibrium position corresponds to vanishing values of $\lambda, \mu, \nu, \ldots$, we make the second supposition that the value of $\varphi(0, 0, 0, \ldots)$ is likewise zero; this is allowed because of the arbitrary constant. If one now determines the constant from the given initial state, for which $v, \lambda, \mu, \nu, \ldots$ are to be denoted by $v_0, \lambda_0, \mu_0, \nu_0, \ldots$, one obtains

$$\Sigma mv^2 = \varphi(\lambda, \mu, \nu, \ldots) - \varphi(\lambda_0, \mu_0, \nu_0, \ldots) + \Sigma mv_0^2.$$

By the assumption that $\varphi(\lambda, \mu, \nu, \ldots)$ has, for $\lambda = 0$, $\mu = 0$, $\nu = 0$, ..., a maximum whose value itself vanishes, one can always specify such small positive quantities $l, m, n, \ldots$ that for every system $\lambda, \mu, \nu, \ldots$, if the numerical values of these variables are respectively no greater than $l, m, n, \ldots$, $\varphi(\lambda, \mu, \nu, \ldots)$ is always negative, except in the single case where $\lambda, \mu, \nu, \ldots$ vanish simultaneously. This case is excluded if we consider only such systems in which at least one of the variables $\lambda, \mu, \nu, \ldots$, apart from sign, is equal to its bound $l, m, n, \ldots$. If, among all these negative values of the function, $-p$ is the smallest apart from sign, it is easy to show that, if $\lambda_0, \mu_0, \nu_0, \ldots$ are taken numerically smaller than $l, m, n, \ldots$ and at the same time the condition

$$-\varphi(\lambda_0, \mu_0, \nu_0, \ldots) + \Sigma mv_0^2 < p$$

is satisfied, then each of the variables $\lambda, \mu, \nu, \ldots$ will remain, in the course of the motion, numerically below its bound $l, m, n, \ldots$. Were the contrary to occur, then, since the initial values $\lambda_0, \mu_0, \nu_0, \ldots$ satisfy the condition just stated by hypothesis, and since the variables $\lambda, \mu, \nu, \ldots$ are continuous,[5] there would have to be a time at which equality first occurs between one or more of the numerical values of $\lambda, \mu, \nu, \ldots$ and the corresponding bound $l, m, n, \ldots$, without any of the remaining variables having yet exceeded its bound. For this instant the negative value of $\varphi(\lambda, \mu, \nu, \ldots)$
would not be numerically smaller than p, and therefore the right-hand side of our equation, in view of the preceding inequality relating to the initial state, would be negative; this contradicts the nature of the left-hand side. At the same time it is clear that the velocities v will also always remain enclosed within bounds, since one always has

$$\Sigma mv^2 \leq \Sigma mv_0^2 - \varphi(\lambda_0, \mu_0, \nu_0, \ldots),$$

[3] Traité de Mécanique, tome second, page 492.

[4] This citation is absent in the Academy report. K.

[5] The words "and since the variables $\lambda, \mu, \nu, \ldots$ are continuous" are absent in the Academy report. K.

and also that the bounds for each v, as well as for each of the variables $\lambda, \mu, \nu, \ldots$ determining the position of the system, can be made arbitrarily narrow, since $l, m, n, \ldots$ are capable of every degree of smallness.[6]

Finally, we wish to point out a remarkable error, found in various writers, concerning the subject just discussed.[7] It is asserted that if the system passes through several equilibrium positions in the course of its motion, then these are alternately positions of stable and unstable equilibrium, that is, positions to which a maximum or a minimum of the function $\varphi(\lambda, \mu, \nu, \ldots)$ corresponds. This assertion is based, first, on the fact that a maximum or minimum does not lose this character when the variables $\lambda, \mu, \nu, \ldots$ formerly regarded as independent obtain the special values for which such a maximum or minimum occurs by becoming functions of a new quantity t; and second, on the fact that for a function of a single variable t, maxima and minima follow one another alternately. Both statements are correct, but they do not justify the conclusion drawn from them. Rather, it would be necessary for the first statement also to remain valid conversely: that is, that the function $\varphi(\lambda, \mu, \nu, \ldots)$ depending only on t, if it presents a maximum or minimum for a particular t, should retain the same property when the corresponding values of $\lambda, \mu, \nu, \ldots$ are regarded as special values of the quantities $\lambda, \mu, \nu, \ldots$ considered as independent variables. This need not be the case, even if only one quantity, for example λ, occurs. Thus the function λ^2 has only a minimum and no maximum at all, whereas the function $\sin^2 t$, into which the former passes under the assumption $\lambda = \sin t$, has not only infinitely many minima, which are equal to one another and to the former one, but also infinitely many maxima. The error into which one has fallen in this respect must be all the more surprising since the incorrectness of the assertion made is already recognizable in the ordinary motion of the pendulum.

[6] Here the extract in the Academy report closes; cf. p. 37 in the 1846 volume. K.

[7] Traité de Mécanique par Poisson, tome second, page 491.

II.

On a General Means of Verifying the Expression for the Potential Relative to an Arbitrary Mass, Homogeneous or Heterogeneous.

By M. G. Lejeune Dirichlet.

Crelle, Journal für die reine und angewandte Mathematik, Bd. 32, p. 80–84.

On a General Means of Verifying the Expression for the Potential Relative to an Arbitrary Mass, Homogeneous or Heterogeneous.

Given an arbitrary mass, bounded in every direction, forming a continuous whole or composed of several separated parts, and having at each of its points a finite density, if one forms the sum of all the elements of this mass, each divided by its distance from an arbitrary point m, one obtains what geometers have agreed to call the potential of the mass with respect to that point. It is known that the triple integral thus defined, which is always a determinate function of the rectangular coordinates x, y, z of the point m, has a great number of important properties. I shall recall here only those on which the observation that is the object of this note bears.

1) The potential v and its first-order differential coefficients

$$\frac{\partial v}{\partial x}, \quad \frac{\partial v}{\partial y}, \quad \frac{\partial v}{\partial z},$$

which express the components of the attraction exerted by the mass at the point m, are finite and continuous functions of x, y, z throughout all of space.

2) There always exist determinate bounds which the products

$$xv, \quad yv, \quad zv, \quad x^2\frac{\partial v}{\partial x}, \quad y^2\frac{\partial v}{\partial y}, \quad z^2\frac{\partial v}{\partial z}$$

cannot exceed at any point of space.

3) If one leaves aside the particular points around which the density varies abruptly, each of the three derivatives

$$\frac{\partial^2 v}{\partial x^2}, \quad \frac{\partial^2 v}{\partial y^2}, \quad \frac{\partial^2 v}{\partial z^2}$$

will always have a finite and unique value, and these simultaneous values will be such that

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = -4\pi\varrho, \tag{a}$$

where ϱ denotes the density at the point (x, y, z), to be considered zero when this point lies outside the mass. For the points just excepted, which will always be situated on certain surfaces, among which are also included those which bound the mass, or at least the parts of the latter where the density is not zero, the functions

$$\frac{\partial^2 v}{\partial x^2}, \quad \frac{\partial^2 v}{\partial y^2}, \quad \frac{\partial^2 v}{\partial z^2}$$

will generally each have two distinct values, but always finite ones; for the first, for example, these will be the two limits of the quotient

$$\frac{\Delta \dfrac{\partial v}{\partial x}}{\Delta x},$$

corresponding to an infinitely small positive or negative value of Δx. From the combination of these double values there will in general result eight different values for the trinomial (a); it is not necessary to consider them. It suffices for our object to know that they are all finite.

Here is now the very simple observation to which the properties just stated give rise, and which, so far as I know, had not yet been made. These properties completely characterize the potential v, and there exists no other function v' possessing all these properties together.

To prove this, let us suppose that the contrary takes place, and see what will result for the difference $v' - v = u$. From the nature of the properties in question, it is evident that the new function u will still satisfy conditions (1) and (2), while equation (a) will be replaced by

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0.$$

The points mentioned above must, however, be excepted; we know only that finite values of the trinomial correspond to such points. These finite values - unknown for the moment, but in fact all equal to zero, since we are going to see that $u = 0$ - occurring only along certain

surfaces, will have no influence on the result of a triple integration, and we shall have rigorously

$$\iiint u \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) dx\, dy\, dz = 0,$$

the limits being arbitrary. Let us extend each of the three integrations from $-a$ to $+a$, and consider in particular the integral of the first term. If we begin with the operation relative to x, integration by parts gives

$$\int_{-a}^{+a} u \frac{\partial^2 u}{\partial x^2}\, dx = \left(u \frac{\partial u}{\partial x} \right)_{-a}^{+a} - \int_{-a}^{+a} \left(\frac{\partial u}{\partial x} \right)^2 dx,$$

the indices appended to the first term indicating that one must take the difference of the two values corresponding to $+a$ and to $-a$. It is indeed to this difference alone that the terms produced outside the sign by the integration by parts reduce, $u \frac{\partial u}{\partial x}$ being a continuous function of x. Proceeding in an analogous manner on the other two integrals, one obtains

$$\begin{aligned} &\iiint \left(\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 + \left(\frac{\partial u}{\partial z} \right)^2 \right) dx\, dy\, dz \\ &= \iint \left(u \frac{\partial u}{\partial x} \right)_{-a}^{+a} dy\, dz + \iint \left(u \frac{\partial u}{\partial y} \right)_{-a}^{+a} dx\, dz + \iint \left(u \frac{\partial u}{\partial z} \right)_{-a}^{+a} dx\, dy. \end{aligned}$$

By virtue of conditions (2), the function of y and z that stands under the sign in the first double integral will, throughout the extent of that integral, be numerically less than $\frac{k}{a^3}$, where k does not depend on a; the integral is therefore less than $\frac{4k}{a}$, and vanishes for $a = \infty$. The same thing holding for the other two, it follows that the triple integral, taken between infinite limits, is likewise zero. But the function subjected to the triple integration, being continuous and never capable of becoming negative, must be identically zero; consequently u has a constant value, which can only be zero, since u must vanish at infinity. This was to be shown.

The result just obtained supplies a general means of verifying the expression for the potential when this potential is given throughout all of space. It will suffice, for this purpose, to make sure that the given expression satisfies all the conditions enumerated above.

The application of this procedure to the known formulae relating to a homogeneous ellipsoid perhaps offers the most expeditious means of establishing the theorems concerning that celebrated question, provided, however, that one seeks only a simple verification and does not have in view a method that naturally leads to the supposedly unknown results at the same time as it proves them.

The equation of the ellipsoid being

$$\frac{x^2}{\alpha^2} + \frac{y^2}{\beta^2} + \frac{z^2}{\gamma^2} = 1,$$

the trinomial forming its left-hand side will be less than or greater than unity according as the point (x, y, z) is inside or outside the ellipsoid. In the second case, the equation

$$\frac{x^2}{\alpha^2+\sigma} + \frac{y^2}{\beta^2+\sigma} + \frac{z^2}{\gamma^2+\sigma} = 1$$

has a unique positive root σ, which varies continuously with the variables x, y and z. This is seen by observing that the derivatives

$$\frac{\partial \sigma}{\partial x}, \quad \frac{\partial \sigma}{\partial y}, \quad \frac{\partial \sigma}{\partial z},$$

given by the equations

$$l\frac{\partial \sigma}{\partial x} = \frac{2x}{\alpha^2+\sigma}, \quad l\frac{\partial \sigma}{\partial y} = \frac{2y}{\beta^2+\sigma}, \quad l\frac{\partial \sigma}{\partial z} = \frac{2z}{\gamma^2+\sigma}, \tag{b}$$

where

$$l = \frac{x^2}{(\alpha^2+\sigma)^2} + \frac{y^2}{(\beta^2+\sigma)^2} + \frac{z^2}{(\gamma^2+\sigma)^2},$$

always have finite values. This being set, if for brevity we put

$$D = \sqrt{\left(1+\frac{s}{\alpha^2}\right)\left(1+\frac{s}{\beta^2}\right)\left(1+\frac{s}{\gamma^2}\right)},$$

it remains to prove that one has

$$v = \pi \int_0^\infty \left(1 - \frac{x^2}{\alpha^2+s} - \frac{y^2}{\beta^2+s} - \frac{z^2}{\gamma^2+s}\right)\frac{ds}{D},$$

or

$$v = \pi \int_\sigma^\infty \left(1 - \frac{x^2}{\alpha^2+s} - \frac{y^2}{\beta^2+s} - \frac{z^2}{\gamma^2+s}\right)\frac{ds}{D},$$

according as the point (x, y, z) is located inside or outside the mass, whose density is assumed equal to unity. Differentiation gives

$$\frac{\partial v}{\partial x} = -2\pi x \int_0^\infty \frac{ds}{(\alpha^2+s)D},$$

or

$$\frac{\partial v}{\partial x} = -2\pi x \int_\sigma^\infty \frac{ds}{(\alpha^2+s)D},$$

the term arising from the variability of the limit σ in the second case vanishing by virtue of the equation of which σ is a root. It is manifest that the first expressions for v and $\frac{\partial v}{\partial x}$, and the analogous expressions for $\frac{\partial v}{\partial y}$, $\frac{\partial v}{\partial z}$, vary continuously with the coordinates x, y, z, and that the same is true of those corresponding to an exterior point; one also sees that continuity is not broken at the surface where $\sigma = 0$, which makes the corresponding expressions coincide.

Conditions (2) plainly hold for an interior point and, in this case, are already included in conditions (1). To verify them outside the ellipsoid, let λ be the smallest of the semiaxes α, β, γ. Observing that the factor $\frac{ds}{D}$ in the second expression for v is less than unity, and attending only to numerical values, we evidently have

$$v < \frac{2\pi\alpha\beta\gamma}{(\lambda^2+\sigma)^{1/2}}, \quad \frac{1}{x}\cdot\frac{\partial v}{\partial x} < \frac{4\pi\alpha\beta\gamma}{3(\lambda^2+\sigma)^{3/2}}.$$

As, on the other hand, the equation in σ gives, apart from the sign, $x < (\alpha^2+\sigma)^{1/2}$, one concludes

$$xv < 2\pi\alpha\beta\gamma\left(\frac{\alpha^2+\sigma}{\lambda^2+\sigma}\right)^{1/2}, \quad x^2\frac{\partial v}{\partial x} < \frac{4}{3}\pi\alpha\beta\gamma\left(\frac{\alpha^2+\sigma}{\lambda^2+\sigma}\right)^{3/2},$$

inequalities whose right-hand sides remain finite as σ increases from zero to infinity.

Let us pass to the derivatives of the second order. One finds

$$\frac{\partial^2 v}{\partial x^2} = -2\pi\int_0^\infty \frac{ds}{(\alpha^2+s)D}$$

or

$$\frac{\partial^2 v}{\partial x^2} = \frac{2\pi x}{(\alpha^2+\sigma)\Delta}\cdot\frac{\partial\sigma}{\partial x} - 2\pi\int_\sigma^\infty \frac{ds}{(\alpha^2+s)D},$$

where Δ denotes the value of D corresponding to $s = \sigma$. These expressions are still finite and determinate, but they no longer coincide at the surface. If we now add to each of these equations the two analogous equations, and observe that one has, indefinitely,

$$\int\left(\frac{1}{\alpha^2+s}+\frac{1}{\beta^2+s}+\frac{1}{\gamma^2+s}\right)\frac{ds}{D} = -\frac{2}{D} + \text{const.},$$

there will come, according to the two cases,

$$\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}+\frac{\partial^2 v}{\partial z^2} = -4\pi,$$

or

$$\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}+\frac{\partial^2 v}{\partial z^2} = \left(\frac{2x}{\alpha^2+\sigma}\cdot\frac{\partial\sigma}{\partial x}+\frac{2y}{\beta^2+\sigma}\cdot\frac{\partial\sigma}{\partial y}+\frac{2z}{\gamma^2+\sigma}\cdot\frac{\partial\sigma}{\partial z}-4\right)\frac{\pi}{\Delta}.$$

The first of these two equations already agrees with condition (a), and to make sure that the same is true of the second it suffices to take the sum of the three equations (b), each multiplied by its right-hand side, and then to suppress the common factor l, which is evidently different from zero. One thus recognizes that the right-hand side of the last equation vanishes, as it must for an exterior point.

We have supposed up to now that the given mass has three dimensions. When this mass has only two and is distributed over one or over several surfaces, the method of verification just set out will have to undergo certain modifications, which nevertheless follow too simply from the known theorems relating to that case for it to be necessary to enter into further details on this point.[1]

[1]Cf. the immediately following paper. K.

III.
On the Characteristic Properties of the Potential of a Mass Distributed over One or More Finite Surfaces.

By G. Lejeune Dirichlet.

Bericht über die Verhandlungen der Königl. Preuss. Akademie der Wissenschaften. Jahrg. 1846, S. 211–212.

On
the Characteristic Properties of the Potential of a Mass Distributed over One or More Finite Surfaces.

As was already observed in an earlier paper,[1] the method developed there for the case of a mass extended in three dimensions can also be applied to the question just mentioned, and it follows that, for a mass distributed over surfaces, the characteristic properties of the potential, that is, those which completely determine its expression, are the following:

1) The potential v is, throughout all of space, a finite and continuous function of the rectangular coordinates x, y, z.

2) The three partial differential quotients of the potential taken with respect to the coordinates are likewise continuous everywhere outside the surfaces. For a point lying on them, however, there is an interruption of continuity consisting in this: if the potential at such a point and at the points lying on the normal on both sides at distance ε are denoted by v, v', v'', then the quotient

$$\frac{v' + v'' - 2v}{\varepsilon}$$

has, for an infinitely small positive ε, the expression $-4\pi\varrho$ as its limit, where ϱ denotes the density at the point of the surface.

3) Everywhere outside the surfaces the equation

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = 0$$

holds.

4) At infinite distance from the surfaces, the same conditions are fulfilled as for a mass extended in three dimensions.

Apart from the utility that may be afforded by the proof that the properties just listed completely characterize the potential, in order to verify an expression for the potential known from wherever it may have been obtained, this proof also presents another and more essential interest. It follows from it, namely, that the important theorem stated by Gauss, according to which a mass can always be distributed over given surfaces in such a way that the potential corresponding to this distribution receives, at every point of the surfaces, an arbitrarily prescribed value varying according to continuity, is wholly identical with the assertion that in a homogeneous mass filling all of space, provided that initially only temperatures vanishing at infinite distance

[1] *Crelle's Journal*, Band 32, p. 80. Editorial note in the Werke: P. 9 of the second volume of this edition of G. Lejeune Dirichlet's works. The observation is found at the end of the paper on p. 16. K.

occur, a fixed temperature will always be established everywhere after an infinite time under the influence of constant heat sources distributed over surfaces. The identity of the two theorems, and the new relation between two branches of mathematical physics already exhibiting such great kinship that is thereby established, is immediately clear from what has been said above, since the known conditions determining the permanent state of heat coincide exactly with those by which the potential corresponding to the required mass distribution is characterized.

IV.
On the Reduction of Positive Quadratic Forms with Three Indeterminate Integers.

By G. Lejeune Dirichlet.

Report on the proceedings of the Royal Prussian Academy of Sciences, year 1848, pp. 285–288.

On
the reduction of positive quadratic forms
with three indeterminate integers.

It is well known that Lagrange was the first to show that every binary quadratic form can be reduced, that is, transformed into another one whose coefficients satisfy certain inequality conditions; he also showed at the same time that in every class of positive forms there always exists only a single such form, so that in this case the various reduced forms corresponding to a given determinant can serve as the representatives of the various classes. After the ternary forms had later been considered in the *Disquisitiones arithmeticae* from a general point of view, it became necessary for the further development of this theory to extend Lagrange's investigation to the positive forms of this kind, that is, to find such inequality conditions among the coefficients that, in every class, they are satisfied by one and only one form. This extension, which involved great difficulties, was accomplished by Seeber in a work devoted especially to the positive ternary forms, of which it forms the principal content. The great complication of Seeber's method made it desirable to find a simpler path leading to the same results; this path, however, could be found only after many fruitless attempts.

For an easier description of this new procedure, which is of surprising simplicity, it will be expedient to clothe the matter in a geometric form,

using here the interesting geometric construction by which Gauss, in a short essay in which he discusses Seeber's work,[1] represents the principal properties of positive binary and ternary forms. According to this geometric representation, every positive ternary form corresponds to an infinite system of points arranged parallelepipedally, and the various subformations of the form are nothing other than changed arrangements of the same system according to another elementary parallelepiped. Expressed in this language, Seeber's results consist essentially in the following.

1. Every system of parallelepipedally arranged points can always be divided in such a way that, in the corresponding elementary parallelepiped, neither the sides of the faces are greater than their diagonals, nor the edges of the parallelepiped greater than the diagonals of the parallelepiped.

2. For a given system, such an arrangement can in general be carried out in only one way.

One is easily convinced of the correctness of the first of these propositions by the following very simple consideration.

Let (0) be any point of the system. The remaining points of the system evidently always occur in pairs at equal distance and in opposite directions from (0). Let (1) be one of the points

[1] *Crelle's Journal*, vol. 20, p. 312. Editorial note in the *Werke*: Gauss' Werke, vol. II, p. 188. K.

in the pair for which the distance from (0) is smaller than for every other pair. If the same least distance occurs for several pairs, choose (1) arbitrarily in any one of them. Now place through the line (01) and any point of the system outside that line an infinite plane; all points falling in this plane then form a parallelogrammatic system. In one of the two nearest parallel lines of this system take the point lying nearest to (0), or arbitrarily one of them if the same shortest distance occurs for two points of this line. Among all planes that can be placed through (01) in the indicated way, this shortest distance will be smaller in one plane, or in some of them, than in all the others. Fix this plane, or one of them if the same shortest distance should be common to several, and the point in it which we shall denote by (2). Then one evidently has:

$$(02) \geq (01),$$

and just as easily one sees that, in the parallelogram determined by the sides (01) and (02), the two diagonals are not smaller than these sides. Once the plane (012) has been fixed in this way, observe that the entire system of points lies in the plane (012) and in other planes parallel to it and mutually equidistant. Now take, in one of the two planes adjacent to the plane (012), the point nearest to (0), or one of the nearest points if the same shortest distance should occur for several. If this point is called (3), then one evidently has:

$$(03) \geq (02),$$

and the fundamental parallelepiped determined by the edges (01), (02), and (03) will have the required properties.

After what was remarked above about the parallelogram with sides (01) and (02), it remains for us to show that, both in the parallelograms (013) and (023) and in the parallelepiped, the sides and edges are not greater than the diagonals. In view of the two inequalities

$$(03) \geq (02) \geq (01)$$

this evidently amounts to proving that (03) is no greater than any one of the four diagonals of the named parallelograms, nor greater than any one of the four diagonals of the parallelepiped. But these eight diagonals evidently coincide, in length, with the eight straight lines that can be drawn from (0) to the eight corners of the parallelograms that lie around (3) in the plane through (3) parallel to (012). That none of these eight joining lines is smaller than (03) follows immediately from the condition by which the point (3) was chosen.

If, in the construction just indicated, (1), (2), and (3) are completely determined points (apart from the possibility, always present, that for each of them one can also take the point that lies, with respect to (0), at equal distance and in the opposite direction), then the edges (01), (02), and (03) of the fundamental parallelepiped obtained are unequal, and the diagonals are actually greater than the sides and edges by which, in general, they are not to be exceeded. In this case one then easily proves

that this system admits only this single elementary parallelepiped with the required properties.

The matter takes a somewhat different form when, in our construction, a choice can be made among different non-opposite positions for the points (1), (2), and (3). In such singular cases there may exist, for the system, several non-congruent fundamental parallelepipeds possessing the required properties. All these parallelepipeds, however, can be completely enumerated without difficulty, and one can always distinguish one of them from the others by certain subsidiary conditions, so that, with the addition of these secondary conditions, the theorem that every class contains only one reduced form does not lose its validity. Something similar is already well known in the theory of positive binary forms, where in two special cases one must prescribe a definite sign for the middle coefficient if one wishes to retain only one reduced form in each class.

V.

On the Reduction of Positive Quadratic Forms with Three Undetermined Integral Numbers.

By G. Lejeune Dirichlet.

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It is well known that Lagrange first showed that every binary quadratic form can be reduced, that is, transformed into an equivalent form whose coefficients satisfy certain inequalities. At the same time he showed that in each class of positive forms there is always only one such form, so that, in this case, the several reduced forms belonging to a given determinant may serve as representatives of the several classes. After the ternary forms had later been treated from a general point of view in the *Disquisitiones arithmeticae*, the further development of this theory required the extension to positive ternary forms of the investigation carried out by Lagrange for the positive binary forms: one had to find inequalities among the coefficients such that in each class they are satisfied by one and only one form.

This extension, beset with great difficulties, was carried out by Seeber in a work devoted specifically to positive ternary forms. It forms the principal content of that work, which Gauss characterized as follows in a highly interesting review[2]:

> Full justice must be done to the spirit of thoroughness with which these matters are treated. If we must regret that this thoroughness is accompanied by great length, perhaps discouraging to some–the solution of the problem taking 41 pages and the proof of the theorem 91 pages–this is not to be taken as a reproach. When a difficult problem or theorem is to be solved or proved, the first step, and one to be acknowledged with due thanks, is that a solution or proof be found at all. The question whether this could have been done in an easier and simpler manner remains idle until the possibility has also been decided by the deed. We therefore consider it premature to dwell here on this question.

The great complication of Seeber's method long ago tempted me to try to found the theory of reduced ternary forms in a simpler way. In now communicating the result of these efforts, I believe that, in the interest of brevity and, if I may so speak, of transparency of exposition, I must retain the geometrical form in which I have conducted the investigation. It is based on the remarkable relations which hold between quadratic forms with two or three elements and certain spatial configurations. I begin by carrying out the indications already given by Gauss, in the review just mentioned, concerning these relations.

[1]An extract from this memoir had already been given in the Monthly Report of the Academy. There the principle of this new treatment of the reduction of positive ternary forms–the consideration of successive minima–was indicated, and the proof of the first of Seeber's two results was carried out completely according to this principle.

[2]Crelle's Journal, vol. 20, p. 312.

§. 1.

The ternary form
$$ax^2 + by^2 + cz^2 + 2a'yz + 2b'xz + 2c'xy = \varphi, \tag{1}$$
in which we regard x, y, z as first, second, and third element, is called positive when φ never becomes negative for real values of these elements. In such a form the coefficients
$$a, \quad b, \quad c$$
are always positive, while the combinations of coefficients
$$a'^2 - bc, \quad b'^2 - ac, \quad c'^2 - ab, \quad aa'^2 + bb'^2 + cc'^2 - abc - 2a'b'c' = -D, \tag{2}$$
of which the last, $-D$, is called the determinant of the form, are negative[3]. By virtue of these conditions there are always three acute or obtuse angles λ, μ, ν, completely determined by the equations
$$\cos\lambda = \frac{a'}{\sqrt{bc}}, \qquad \cos\mu = \frac{b'}{\sqrt{ac}}, \qquad \cos\nu = \frac{c'}{\sqrt{ab}},$$
from which a trihedral angle can be formed, since the required condition
$$\cos^2\lambda + \cos^2\mu + \cos^2\nu - 2\cos\lambda\cos\mu\cos\nu < 1$$
coincides with $D > 0$.

Since, however, two mutually symmetric trihedral angles can be formed with the same angles λ, μ, ν, we agree always to choose that one of the two in which the edges, as they lie opposite these angles in order, follow one another from right to left with respect to a straight line drawn from the vertex 0 into the interior of the angle and thought of as going upward. If we now regard the three edges as the positive axes of a coordinate system, the whole infinite space can be referred to our form by taking the products $x\sqrt{a}$, $y\sqrt{b}$, $z\sqrt{c}$ as the coordinates of an arbitrary point. Then φ expresses the square of the distance of this point from the vertex, or, more generally, the square of the distance between two points whose like-named coordinates have those products as differences.

Now form, with three new undetermined elements x', y', z', the linear expressions
$$x = \alpha x' + \beta y' + \gamma z', \quad y = \alpha' x' + \beta' y' + \gamma' z', \quad z = \alpha'' x' + \beta'' y' + \gamma'' z', \tag{3}$$
where the only restriction is that the determinant formed from the nine coefficients
$$\alpha\beta'\gamma'' + \beta\gamma'\alpha'' + \gamma\alpha'\beta'' - \gamma\beta'\alpha'' - \alpha\gamma'\beta'' - \beta\alpha'\gamma'' = E \tag{4}$$
be not zero. Then φ goes over into a new form φ', and the corresponding quantities for that form will be denoted by accented letters[4].

If one again lets an infinite space correspond to the new form, the two infinite spaces are thereby put in pointwise correspondence: two points correspond when, in the expressions for their coordinates
$$x\sqrt{a},\ y\sqrt{b},\ z\sqrt{c}; \qquad x'\sqrt{a'},\ y'\sqrt{b'},\ z'\sqrt{c'},$$
the elements x, y, z and x', y', z' are connected by equations (3). If these expressions are the coordinate differences for two pairs of corresponding points, then the same relations among $x, y, \ldots$ hold as before; hence, by what has just been said and because $\varphi = \varphi'$, the distance

[3] *Disquisitiones arithmeticae*, art. 271.

[4] This is the meaning determined here for the accented letters in this paragraph; it is therefore essentially different from the meaning of a', b', c' in the first display of §. 1. K.

between any two points of the one space is equal to the distance between the corresponding two points of the other. The two pointwise corresponding spaces are therefore either congruent or symmetric: when the origins 0 and $0'$ are made to coincide, either every point falls on its corresponding point or on the opposite point of the latter, where, for brevity, two points of the same space which lie at equal distance and in opposite directions from the origin are called opposite points.

To decide which of these two cases occurs, one draws from the vertex in one space to three arbitrary points and examines whether the straight lines drawn in the other space from its vertex to the corresponding points have the same or the opposite order. For example, in the second space take the lines to the points with coordinates

$$\sqrt{a'}, 0, 0; \qquad 0, \sqrt{b'}, 0; \qquad 0, 0, \sqrt{c'},$$

lying on the positive axes. By the convention above these follow one another from right to left. For the corresponding points in the first space the coordinates are

$$\alpha\sqrt{a'}, \alpha'\sqrt{a'}, \alpha''\sqrt{a'}; \quad \beta\sqrt{b'}, \beta'\sqrt{b'}, \beta''\sqrt{b'}; \quad \gamma\sqrt{c'}, \gamma'\sqrt{c'}, \gamma''\sqrt{c'}.$$

Their order is the same or the opposite according as the determinant E is positive or negative. Thus the transformation is a motion when $E > 0$ and a motion followed by reflection when $E < 0$.

If the elements x, y, z had arbitrary values up to now, let them henceforth be restricted to integers. In place of the whole space we then have an infinite system of parallelepipedally arranged points, that is, a system formed by the intersections of three families of parallel equidistant planes. Suppose now that the substitution coefficients $\alpha, \beta, \gamma, \ldots$ are also integers and that E has the value ± 1. Then to each integral combination x, y, z there corresponds an integral combination x', y', z', and conversely. By what has just been shown, the parallelepipedal systems so put in correspondence can be brought into such a position that one coincides either with the other or with the opposite points of the other. Since, however, the opposite points of such a system again form the same system, the two cases are not distinct. This is also clear from the fact that φ' is unchanged when one changes the signs of $\alpha, \beta, \gamma, \ldots$, which changes E into $-E$.

Thus the two systems are always congruent. Systems corresponding to two equivalent ternary forms φ and φ' are therefore the same spatial configuration in two different arrangements. Conversely, any two different parallelepipedal arrangements of the same system correspond to equivalent forms. Indeed, choose any point of the system as common origin. Between the coordinates relative to the two systems of axes, and hence also between the proportional elements x, y, z, x', y', z', there are linear equations without constant term, that is, equations of the form (3). Since, by assumption, integral values of x, y, z are accompanied by integral values of x', y', z' and conversely, the coefficients $\alpha, \beta, \gamma, \ldots$ are also integers and $E = \pm 1$. On the other hand, for corresponding integral values of the elements one has $\varphi = \varphi'$, and the equality therefore holds identically.

Analogous relations hold between a positive binary form

$$lx^2 + 2mxy + ny^2$$

and a system of parallelogrammatically arranged points. Here take two axes inclined to one another by the angle ϑ determined by

$$m = \sqrt{ln}\cos\vartheta,$$

where the distinction between the axes is made uniformly, for instance by choosing the second to the left of the first after one side of the plane has been designated as the upper side. If $x\sqrt{l}$, $y\sqrt{n}$

are then regarded as coordinates, one obtains a system of points completely determined by the quadratic form, which may be regarded as the intersections of two families of equidistant parallel lines. If between two forms there is the so-called proper equivalence, so that $\alpha\delta - \beta\gamma$ in the substitutions $x = \alpha x' + \beta y'$, $y = \gamma x' + \delta y'$ equals positive unity, then the corresponding systems may be brought to coincide by a motion in the plane. In the other case, where $\alpha\delta - \beta\gamma = -1$, generally speaking one of the systems must be turned over for this purpose.

§. 2.

Having established the connection between quadratic forms and certain geometrical configurations, we must develop some further properties of these configurations. For brevity, a system of parallelogrammatically or parallelepipedally arranged points will be called a *system of second* or *third order*, while an infinite row of equidistant points in a straight line will be called a *system of first order*.

The common character of all three kinds of systems is plainly this: if such a system is brought by a motion without rotation, which we shall call a displacement, into a new position such that one of its points passes into the place originally occupied by another, then the same happens for all points; the system in its new position coincides completely with the system in its original position. It is easy to prove that this displacement property completely characterizes all three kinds of systems: a system with this property is respectively a system of first, second, or third order according as it lies in a straight line and contains two points, lies in a plane and contains three non-collinear points, or contains at least four points not lying in one plane.

For example, suppose all points of a system lie on the same straight line, and let a and a' be two neighbouring points of it. By a displacement carrying a to a', the point a' goes to a point a'' which is as far from a' as a' is from a. Thus a'' also belongs to the system, and the system has no point between a' and a'', since such a point would before the motion have lain between a and a'. Continuing this consideration indefinitely in both directions proves the assertion.

Now suppose a planar system has the displacement property, and let a and a' be two neighbouring points such that no point of the system lies between them on the line aa'. Since the displacement from a to a' carries the infinite line aa' into itself, all points of the system on this line form a system of first order $\ldots a''aa'a'' \ldots$. Since, by hypothesis, the system has at least one further point outside this line, let b be one of the points nearest to the line. If now a displacement carries a to b, the first-order system passes into the new position $\ldots b''bb'b'' \ldots$ and in that position belongs to the original system. It is also clear that neither between the points $\ldots b'', b, b', b'', \ldots$ nor between the lines $\ldots bbb' \ldots$, $\ldots aaa' \ldots$ can any point of the system lie. Continuing in this way, one sees that the whole system can be arranged parallelogrammatically, and that $aa'bb'$ may be chosen as one of its fundamental parallelograms. This construction also gives all parallelogrammatic arrangements of which the system is capable: a' may be chosen arbitrarily subject only to the necessary restriction that no point lie between a and a', and then b may be chosen arbitrarily in the nearest parallel line.

Finally, suppose a system with the displacement property contains at least four points not lying in the same plane. Through any three non-collinear points of it draw a plane. Since every displacement parallel to this plane moves the plane into itself, the points lying in it form, by the preceding result, a system of second order. After this planar system has been divided parallelogrammatically in any way, take one of the remaining points of the spatial system nearest to the plane and displace the system so that an arbitrary point in the plane goes to the chosen point outside it. Repeated application of this motion and of its inverse evidently gives a parallelepipedal arrangement of the given system. At the same time the construction plainly has the required generality, since the first plane, the arrangement of the system of second order in it, and finally the choice of the point in the nearest plane may all be made arbitrarily.

At the end of this paragraph we also show that however one divides the same system of second or third order, the parallelogram or parallelepiped underlying the division always has the same content. This is the geometrical meaning of the theorem that equivalent forms have equal determinants. Namely, in the plane of a system of second order think of a closed curve, for instance a circle. Let z denote the area enclosed by it and s the number of points inside it; it is immaterial whether points on the circumference are counted or not. The quotient z/s, as the radius increases, clearly has the content of a fundamental parallelogram as its limit. Since s and z are independent of the kind of arrangement of the system, the theorem follows for systems of second order. The assertion for spatial systems follows in exactly the same way.

§. 3.

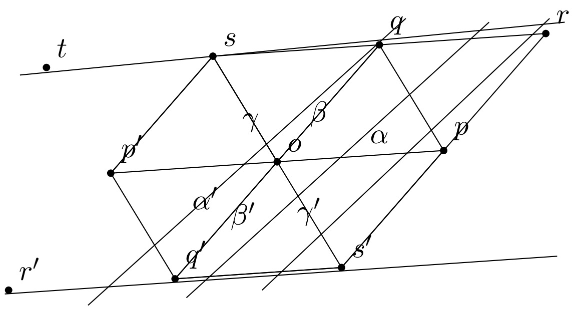
We now show that every system of second order can be divided according to a fundamental parallelogram whose sides are not greater than its diagonals.

I. Let o be any point of the system. The remaining points occur in pairs equidistant from o and in opposite directions. Let p be one of the pair for which the distance from o is smaller than for every other pair; if this shortest distance occurs for more than one pair, choose one of them arbitrarily.

The given system consists of infinitely many mutually congruent and equidistant systems of first order, one of which contains o and p. In one of the two adjacent systems take the point q nearest to o, or, if the same shortest distance occurs for two points, either one. The parallelogram $poqr$ so obtained has the required properties, since by construction

$$op \leq oq, \qquad oq \leq or, \qquad oq \leq os = pq.$$

A fundamental parallelogram satisfying these conditions will henceforth be called *reduced*.



II. We now establish the relation between such a parallelogram and the planar system to which it belongs. If $poqr$ is reduced, then without loss of generality the angle poq may be assumed not obtuse; if it were obtuse, the adjacent parallelogram belonging to the same arrangement would have an acute angle at o. Similarly we may assume $op \leq oq$. Then $or > oq$, and it remains only to consider the condition $pq \geq oq$. Put $op = \sqrt{l}$, $oq = \sqrt{n}$, so that $l \leq n$. Then the relation to the whole system is this: the minimum distance of any point of the system from o is $\sqrt{l}$, and after a point at this distance has been chosen, the second minimum, in all remaining directions outside the line through o and that first point, is $\sqrt{n}$.

This statement is general. The further assertion that the first minimum is attained only at p (opposite points being counted only once), and the second only at q, has the following exceptions:

1°. If $op < oq$ and $oq = pq = os$, the first minimum occurs only at p, the second at q and s.

2°. If $op = oq$ and $oq < pq = os$, the minima are equal, and p and q may be interchanged.

3°. If $op = oq = pq = os$, one of p, q, s may be chosen as first point, and then either of the remaining two as second.

To prove this it is enough, since opposite points are always equally distant from o, to show that q is nearer to o first than all other points on the line sqr, except s, whose distance from o equals $os = pq \geq oq$, and secondly than all points on the following parallel lines.

Since $pq \geq op$, $pq \geq oq$, and the angle poq is not obtuse, the triangle opq, and hence also the congruent triangle oqs, has no obtuse angle. Thus the perpendicular from o to qs falls between s and q (inclusive), proving the first point. Write

$$\cos poq = \frac{m}{\sqrt{ln}},$$

with $m \geq 0$. Then

$$pq^2 = l - 2m + n \geq oq^2 = n,$$

and hence

$$2m \leq l, \qquad 2m \leq n, \qquad 4m^2 \leq ln.$$

If the square of the height of the parallelogram (with $op = \sqrt{l}$ as base) is k, then the square Δ of its area is

$$\Delta = lk = ln - m^2 \geq \frac{3}{4} ln,$$

so that

$$\sqrt{k} \geq \frac{1}{2}\sqrt{3n}.$$

Thus already the second parallel line is at least $\sqrt{3n} = oq\sqrt{3}$ away, proving the second point.

III. Since the successive minima $\sqrt{l}$, $\sqrt{n}$ are determined by the system itself and independently of any particular arrangement, while by what has just been shown they agree in magnitude with the sides of a reduced parallelogram, it follows that if the system admits several such arrangements, the sides of the reduced parallelograms always keep the values $\sqrt{l}$ and $\sqrt{n}$.

All possible fundamental parallelograms are therefore obtained by drawing from o lines to all nearest points (again excluding opposite points), and then taking, in the corresponding nearest parallel line, the nearest point or the two nearest points. By II, these nearest points are nearer to o than all points of the following parallel lines; hence one may omit the separate requirement that the second points be in the first parallel line. Thus all possible arrangements of the system arise by connecting o successively with all pairs of points for which the successive minima occur. Taking account of II, there is in general, and in the second singular case mentioned there, only one arrangement; in the first and third exceptional cases there are respectively two and three.

In the present notation those singular cases correspond to

$$2m = l < n, \qquad 2m < l = n, \qquad 2m = l = n.$$

§. 4.

We have so far considered only properties of the geometrical configurations which may be regarded as constructive representations of known results from the theory of forms. We now solve a different problem: given a system of second order and a fixed point o of it, determine the part of the plane in which every point is nearer to o than to any other point of the system. Since the condition that a point is not farther from o than from another point v is that it lie on the same side as o of the perpendicular erected at the midpoint of ov, we must combine o with all other points and construct the convex polygon formed by the corresponding perpendiculars. Of these infinitely many perpendiculars only finitely many matter; the others do not meet the polygon determined by these. We keep the previous assumptions, so that in the reduced parallelogram $(poqr)$ one has $op \leq oq$, the angle poq is not obtuse, and the angles opq, oqp are acute.

Under these assumptions, one easily proves that only the six vertices p, q, s, p', q', s' of the four parallelograms meeting at o need be considered, and that even the perpendiculars corresponding to s and s' merely touch the desired figure in the special case where poq is a right angle; in that case the same is true of those corresponding to r and r'.

Draw the lines $pq, os, p'q', os'$. One obtains the congruent triangles

$$poq, \quad qos, \quad sop', \quad p'oq, \quad q'os, \quad s'op.$$

If only p, q, s, p', q', s' are considered, one erects perpendiculars at the midpoints of the lines from o to those points, that is, one performs the same construction as if one wanted the circumcentres of the triangles just listed. Since no angle in these triangles is obtuse, two consecutive perpendiculars do not meet outside the corresponding triangle. One obtains the hexagon $\alpha\beta\gamma\alpha'\beta'\gamma'$, with centre o and equal opposite angles and sides, as the region in which every point is nearer to o than to any one of the points p, q, s, p', q', s'. One readily verifies that, except for r and r', the perpendiculars corresponding to the remaining points do not meet this hexagon. By symmetry it suffices to prove this for points in and above the line pop'. For points on the line this is clear; for the points above it, it follows because their distance from o is greater than the diameter of the circle circumscribed about the hexagon. If ϱ denotes the square of its radius, then

$$4\varrho\Delta = ln(l - 2m + n),$$

and from $2m \leq l$, $2m \leq n$, $\Delta \geq \frac{3}{4}ln$ follows

$$4\varrho \leq \frac{4}{3}(l - 2m + n) \leq \frac{8}{3}n.$$

Since the square of the distance from o of points of the second and following parallel lines is at least $3n$, it remains only to consider the points in $tsqr$ other than s, q, r. None of these is nearer to o than t, for which the square of the distance is $4l - 4m + n$. That this is greater than 4ϱ is seen at once after multiplication by Δ and use of $2m \leq l \leq n$[5]. For the point r one likewise sees that the square of its distance from o is $l + 2m + n > 4\varrho$, except in the single case $m = 0$, where the corresponding perpendicular only touches.

It is thus proved that a point lies inside the hexagon $\alpha\beta\gamma\alpha'\beta'\gamma'$, and only then, precisely when it is nearer to o than to any other point of the system. On each side, the distance from o becomes equal to the distance from a second point–for example, for $\alpha\beta$ this point is q–and each vertex of the figure is equally distant from o and from two further points of the system. This last statement changes only in the special case where poq is a right angle; then β and γ, and likewise β' and γ', coincide, and the hexagon becomes a rectangle whose vertices are equidistant from o and from three further points of the system.

It is clear that the same hexagon is obtained whichever reduced parallelogram is used in the singular cases where more than one exists. Likewise, the hexagons or quadrilaterals corresponding to all points of the system are congruent and cover the whole plane.

We also note that the expression

$$\varrho = \frac{ln(l - 2m + n)}{4(ln - m^2)}$$

decreases when, with l and n held constant, m increases from zero to its limiting value $\frac{1}{2}l$. Hence

$$\varrho \leq \frac{1}{4}(l + n) \leq \frac{1}{2}n. \tag{5}$$

[5] The difference $4l - 4m + n - 4\varrho$, multiplied by Δ, can be written as $4m^3 + n(l^2 - m^2) + ln(l - 2m) + l(ln - 4m^2)$, in which the first two terms are positive and the last two non-negative. K.

There is also the inequality

$$2\Delta(n-\varrho) \geq ln^2, \tag{6}$$

whose truth follows immediately after multiplying by 2, bringing all terms to one side, and substituting $\Delta = ln - m^2$, $4\Delta\varrho = ln(l-2m+n)$, whereby it becomes

$$ln(l-2m) + 2mn(n-l) \geq 0.$$

§. 5.

We now come to the proper subject. We must show that every system of third order can be arranged according to a parallelepiped whose faces are reduced parallelograms, and whose edges, four by four equal, do not exceed its diagonals.

Having fixed an arbitrary point (0) of the system, choose in the pair of opposite points for which the distance from (0) is a minimum, or if this minimum occurs for several pairs, arbitrarily in one of these pairs, a point (1).

Among all points outside the line (01) choose again one of the two nearest, call it (2), with arbitrary choice if the same shortest distance occurs for several pairs. Since in the whole system, apart from points on (01), no point is nearer to (0) than (2), the same holds in the plane (102); for the system contained in that plane, (102) is a reduced parallelogram (§. 3, III.). Now take in one of the two nearest parallel planes the point nearest to (0), or, if the minimum occurs more than once, one of the nearest, and join (0) to the chosen point (3). The parallelepiped with edges (01), (02), (03) will then satisfy the requirement. First, by construction,

$$(01) \leq (02) \leq (03).$$

For the base faces of the parallelepiped–that is, the opposite faces containing the two edges that do not exceed the third in size; the other four faces will be called side faces–we already know that they are reduced. By the double inequality just noted we have only to show that the four diagonals of the side faces and the four body diagonals are not smaller than (03). These eight diagonals agree in length with the eight lines from (0) to the eight points lying around (3) in the plane of the upper base face, the eight vertices of the four parallelograms meeting at (3). None of these connecting lines is smaller than (03), by the condition under which (3) was chosen.

Having seen that a system of third order can always be divided according to a reduced parallelepiped, we now establish the relation between such a parallelepiped and the system, and especially compare the distances of the points of the system from (0). Put

$$(01) = \sqrt{a}, \qquad (02) = \sqrt{b}, \qquad (03) = \sqrt{c},$$

and keep throughout the assumption $a \leq b \leq c$.

1°. In the plane of the base face, the relations discussed above (§. 3, II.) hold; hence the successive minima of distance have magnitudes $\sqrt{a}$, $\sqrt{b}$, with the possible freedom in the singular cases described there.

2°. Consider now the points outside the plane of the lower base face, first those in the plane of the upper base face. Since, by the assumption that the parallelepiped is reduced, the line (03) is not greater than any line drawn from (0) to the eight points lying around (3), the foot of the perpendicular from (0) to the plane of the upper base face is not farther from (3) than from any of those eight points. Thus this foot does not fall outside the hexagon or quadrilateral belonging to (3) that was constructed in the preceding paragraph. Among the eight points, exceptionally, if the foot falls on a side, one point, or if the foot coincides with a vertex, two points (three when the polygon becomes a rectangle), may be as near to the foot as (3) is; all other points of the plane are farther away from it. It follows that the shortest distance from (0) to a point of the upper base face, equal to $\sqrt{c}$, occurs generally only for (3), but exceptionally may occur for one, two, or even three further points.

3°. For the following parallel planes we need a bound for the square h of the perpendicular just mentioned. Since its foot does not fall outside the hexagon belonging to (3), if ϱ denotes the square of the radius of the circumscribed circle, then

$$h \geq c - \varrho.$$

But by §. 4 also $\varrho \leq \frac{1}{2}b \leq \frac{1}{2}c$, hence $h \geq \frac{1}{2}c$. Thus already the second parallel plane is at least $\sqrt{2c}$ distant, so above the upper base face there are only points whose distance from (0) is greater than $\sqrt{c}$.

Summing up, the minimum distance in the whole system has value $\sqrt{a}$; after a point at this distance has been chosen, the minimum in the remaining directions is $\sqrt{b}$; and after the second point has also been fixed, the least distance from (0) to points outside the plane determined by (0) and the two first points is $\sqrt{c}$. The successive minima $\sqrt{a}$, $\sqrt{b}$, $\sqrt{c}$ are therefore completely determined in magnitude, though not always in position. The possible exceptions are exactly those arising from the singular cases in the base plane, together with the exceptional positions in the upper base plane described above.

Consequently every reduced arrangement of the system arises by choosing, in succession, points at which these successive minima are attained. In the ordinary case the arrangement is unique. If equalities among the minima or among the boundary conditions occur, several reduced parallelepipeds may appear; but they are precisely those obtained by the stated alternative choices of the points realizing the successive minima.

It follows in particular that if a reduced parallelepiped is such that every segment which is required not to exceed certain others is in fact strictly smaller than them, then the corresponding system has no second reduced arrangement. Conversely, a second reduced arrangement can occur only when one of the limiting equalities holds.

Thus, when a system admits more than one reduced parallelepiped, knowledge of one arrangement suffices to find all the others. The first, non-singular case occurs always and only when the reduced parallelepiped given by this arrangement is of such a nature that all the lines which must not be exceeded by others are in fact strictly smaller: all face diagonals are greater than the corresponding sides, and all body diagonals of the parallelepiped are greater than its edges.

§. 6.

We now translate the results of the preceding paragraph to ternary forms. For uniformity, and to avoid useless distinctions, suppose that by interchanging or changing the signs of the undetermined elements–which does not remove the form from its class–each ternary form

$$ax^2 + by^2 + cz^2 + 2a'yz + 2b'xz + 2c'xy \tag{7}$$

has first been put into a shape such that $a \leq b \leq c$; secondly, among the coefficients a', b', c', unless all three are non-zero and negative, none has the negative sign; and thirdly, if $b = c$, then c', disregarding its sign, is not greater than b', if $a = b$, then b' is not greater than a', and if $a = b = c$, then neither c' is greater than b' nor b' greater than a'. It is easy to see that these requirements can be satisfied in only one way. Their introduction has the advantage that, just as before every ternary form corresponded to a completely determined parallelepiped, now to every parallelepiped there belongs an analytic expression whose coefficients are completely determined also in their order and in their signs.

Under this convention we call the form (1), for which $a \le b \le c$, *reduced* when it corresponds to a reduced parallelepiped. Since the diagonals of the faces must not be smaller than the sides of those faces, one has

$$a \pm 2c' + b \ge b, \qquad a \pm 2b' + c \ge c, \qquad b \pm 2a' + c \ge c.$$

Put $\sigma = -1$ when a', b', c' are all three negative, and otherwise $\sigma = 1$. These conditions are equivalent to

$$a \ge 2c'\sigma, \qquad a \ge 2b'\sigma, \qquad b \ge 2a'\sigma, \tag{8}$$

and only when equality occurs will one of the diagonals in the corresponding parallelogram be equal to a side. The conditions concerning the body diagonals give

$$a + b + c + 2a'\delta + 2b'\varepsilon + 2c'\delta\varepsilon \ge c \qquad (\delta = \pm 1,\ \varepsilon = \pm 1),$$

where the signs in $\delta = \pm 1$, $\varepsilon = \pm 1$ are arbitrary.

First take the case in which none of a', b', c' is negative. Considering the four sign combinations and the condition that, if a and b are equal, then $b' \le a'$, one sees at once that the preceding inequality is then automatically satisfied by virtue of the conditions already obtained. The limiting case of equality, in which a body diagonal becomes equal to the edge $\sqrt{c}$, can occur only once, and only when one of b', c' is zero and when, at the same time, among the conditions (2), both the one involving the other of the two quantities b', c' and the one involving a' are limiting equalities. If a', b', c' are negative, the inequality is always satisfied, and without a limiting equality except when $\delta = \varepsilon = 1$. Thus one must add the new condition

$$a + b + 2a' + 2b' + 2c' \ge 0, \tag{9}$$

where again the lower sign refers to a diagonal becoming equal to the edge $\sqrt{c}$.

If the conditions (2), and also (3) when a', b', c' are negative, are satisfied without equality in any of the inequalities, then in the class to which the form belongs there is no second distinct reduced form, whether with or without equality in the defining conditions. This follows from the end of the preceding paragraph: the corresponding point system can then be divided according to only one reduced parallelepiped. The situation is different when not all conditions hold with strict inequality. Then several reduced forms may occur in the same class, and they can be derived from a given one. It suffices to show this in a principal case; we choose the case $b < c$.

Assume first $a > 2c'\sigma$. Then only the direction of the edge $\sqrt{c}$ can be changed, namely if in the plane of the upper base face there are one or more further points whose distance from the vertex is $\sqrt{c}$. If $\xi, \eta, 1$ are the values of the elements corresponding to such a point, then, when the third edge is directed toward it, all coefficients remain unchanged except a', b', which become respectively

$$a' + c'\xi + b\eta, \qquad b' + a\xi + c'\eta,$$

as is seen easily and almost without calculation.

By the preceding enumeration, the values of ξ, η satisfying the condition are, when $a = 2b'\sigma$,

$$\xi = -\sigma, \qquad \eta = 0;$$

when $b = 2a'\sigma$,

$$\xi = 0, \qquad \eta = -\sigma;$$

when simultaneously $a = 2b'$, $b = 2a'$, $c' = 0$,

$$\xi = -1, \qquad \eta = -1;$$

and when a', b', c' are negative and satisfy

$$a + b + 2a' + 2b' + 2c' = 0,$$

then

$$\xi = 1, \qquad \eta = 1.$$

Corresponding to these four hypotheses, a', b' are transformed into

$$a' - c'\sigma,\ b' - a\sigma (= -b'); \qquad -a',\ b' - c'\sigma; \qquad -a',\ -b'; \qquad a' + b + c',\ a + b' + c'.$$

The third case, and in general the assumption $c' = 0$, may be disregarded, since it gives a new form which, after the prescribed changes of sign at the beginning of this paragraph, is plainly identical with the original one. In each of the three remaining cases one obtains, after making the necessary sign changes, a new reduced form in the same class, unless it coincides with the original. If two of the hypotheses hold simultaneously, two such forms are obtained. This completes the enumeration, since all three hypotheses clearly cannot hold simultaneously. If, still under the assumption $b < c$, one had $a = 2c'\sigma$, then, if $a < b$, the second edge in the base face would have had to be turned; if $a = b$, the first edge could also have been brought into the original position of the second; and this new position, or these two new positions, of the first two edges would have had to be combined with all directions of the third, including the original one.

The inconvenience that in singular cases several reduced forms may occur in the same class can easily be removed by adding to the general definition certain secondary conditions for such cases. They arise, for example, if one requires that the last coefficient c', when it is not completely determined, have the smallest numerical value of which it is capable among the reduced forms of the class, and then proceeds similarly with b'. To illustrate this, consider among the singular cases just treated the case $b < c$, where none of the three conditions (2) holds with the lower sign, but the three negative

values a', b', c' satisfy

$$a + b + 2a' + 2b' + 2c' = 0.$$

As noted above, c' is then determined, and there are only two reduced forms. If a' and b' are the values of the fourth and fifth coefficients in one of them, then in the other they are $a' + b + c'$, $a + b' + c'$, or, since these latter values are evidently positive while c' is negative and therefore z must be replaced by $-z$ to satisfy the sign convention, rather

$$-(a' + b + c'), \qquad -(a + b' + c').$$

As is natural, substituting these values for a', b' again gives the equation

$$a + b + 2a' + 2b' + 2c' = 0,$$

and the original values a', b' arise from them in the same way as they arose from a', b'. Thus the fifth coefficient can have only the two negative values b' and $-(a + b' + c')$, whose sum is $-a - c'$. Hence, if one adds to the defining conditions the further condition

$$-b' \leq \frac{1}{2}(a + c'),$$

the class will contain only one reduced form.

To conclude the memoir we derive from our principles a beautiful theorem found by Seeber by induction and proved by Gauss in the often mentioned review. According to this theorem, in a reduced form the product of the first three coefficients is not greater than twice the absolute value of the determinant.

Since the absolute value of the determinant is equal to the square of the volume of the parallelepiped corresponding to the form, the inequality to be proved, in the notation used in §. 5, 3°, is

$$abc \leq 2\Delta h,$$

where Δ denotes the square of the base area. Put

$$c = b + t,$$

where $t \geq 0$. From the inequality obtained in §. 5, 3°,

$$h \geq c - \varrho = b - \varrho + t,$$

after multiplying it by 2Δ, subtract the equation

$$abc = ab^2 + abt.$$

This gives

$$2\Delta h - abc \geq 2\Delta(b - \varrho) - ab^2 + (2\Delta - ab)t.$$

But by the inequality proved at the end of §. 4, $2\Delta(b-\varrho)-ab^2$ is non-negative, and $2\Delta - ab \geq \frac{1}{2}ab$ is positive. The theorem follows.

VI.

On the Determination of Mean Values in Number Theory

G. Lejeune Dirichlet

Proceedings of the Royal Prussian Academy, 1849, pp. 69–83.

On the determination of mean values in number theory.

[Read in the Academy of Sciences on 9 August 1849[1].]

Although the functions considered in number theory are almost never representable by analytic expressions and appear to proceed quite irregularly, a greater regularity nevertheless emerges in their mean values the further one follows their sequence. That is, there are certain simple expressions which represent the progress of the mean values with continuously increasing accuracy, just as one curve approaches the course of another curve of which it is the asymptote. Toward the end of the fifth section of the *Disquisitiones arithmeticae* one finds several highly remarkable expressions of this kind, relating to the theory of quadratic forms. Since these interesting results, which are only communicated there incidentally and without proof, have not yet been proved, and since in general no methods have been known for treating similar questions, I began several years ago to look for suitable means. From that earlier work, however, I gave nothing to the public except a few new results[2], because there appeared to me the prospect of essentially simplifying the treatment of such problems by continued efforts, and in particular of making it independent of the integral calculus. Other investigations then drew me away from this subject for a longer time. After later taking it up again, I became convinced that in many cases one arrives at the asymptotic expression for the mean value by quite elementary considerations founded on a very simple transformation of series. For the moment I restrict myself to a series of problems for which the indicated means alone is sufficient. In a subsequent paper I shall concern myself with more difficult problems whose solution requires combining the transformation just mentioned with other auxiliary means.

1.

To put the transformation on which the solution of the problems treated in this paper chiefly rests in its proper light, it seems appropriate to begin at once with one of the simplest questions, to carry its solution as far as can be done without that transformation, and then to complete it with its help.

Let $f(n)$ denote the number of divisors of the integer n, and consider the problem of determining the so-called summatory term of this function, that is, the sum

$$f(1) + f(2) + \cdots + f(n) = F(n).$$

If s is an integer $\leq n$, then in as many terms of our sum there will be a unit corresponding to the divisor s as there are multiples of s among the numbers $1, 2, \ldots, n$. But the number of

[1] The corresponding notice in the Academy report of 1849, p. 218, reads: "Mr. Dirichlet read on the determination of mean values in number theory." K.

[2] Cf. pp. 351 and 372 of volume I of this edition of G. Lejeune Dirichlet's works. K.

these multiples is $[n/s]$, if, as everywhere in what follows, we use square brackets to denote the greatest integer contained in the bracketed value. It follows that

$$F(n) = \sum_1^n \left[\frac{n}{s}\right],$$

where the summation sign refers to s. This equation immediately gives an approximate determination, that is, an asymptotic expression for $F(n)$. Since n/s exceeds the integer $[n/s]$ by less than one unit, one has, up to an error that cannot exceed n,

$$F(n) = n\sum_1^n \frac{1}{s}.$$

But

$$\sum_1^n \frac{1}{s} = \log n + C + \frac{1}{2n} + \cdots,$$

where $C = 0.5772156\ldots$. Since, however, the last expression for $F(n)$ is already affected by an error of order n, only the first term of the infinite series is to be retained; and one sees that the equation

$$F(n) = n\log n$$

is exact only up to an error of first order, for which, incidentally, one can easily give a bound that it cannot exceed. Whether the function, which is of order $n\log n$, contains in its asymptotic expression a term of order n with constant coefficient, or, in other words, whether

$$\frac{1}{n}F(n) - \log n$$

approaches a fixed limit as n increases, cannot be decided in this way.

2.

The transformation already mentioned, with which we now wish to occupy ourselves, rests on the simple observation that, while the terms of the sequence

$$\left[\frac{n}{1}\right], \quad \left[\frac{n}{2}\right], \quad \ldots, \quad \left[\frac{n}{s}\right], \quad \ldots,$$

which stops with the nth term, at first decrease very rapidly, from a certain point on each term is either equal to the following one or exceeds it by one unit. This will plainly be the case as soon as the difference

$$\frac{n}{s} - \frac{n}{s+1} = \frac{n}{s(s+1)}$$

has become equal to one or to a proper fraction. Thus if one determines the least integer μ satisfying

$$\mu(\mu+1) \geq n,$$

the mentioned property will occur at the latest from the μth term inclusive. Since for our purpose it is quite immaterial whether one more or one fewer term is drawn into the intended transformation, we shall, for greater simplicity, choose for μ the integer that is either equal to $\sqrt{n}$ or, when this is irrational, lies immediately above it. Thus

$$\mu^2 \geq n,$$

and consequently $\mu(\mu+1) > n$. Put

$$\left[\frac{n}{\mu}\right] = \nu.$$

Then, as is easy to see, ν can differ from μ by at most two units. Let t be any of the numbers $\nu, \nu-1, \dots, 2, 1$. It is then easy to determine the index s of the term furthest from the beginning for which $[n/s]$ has the value t, and after which a term with value $t-1$ follows. The desired index is given by the double condition

$$\frac{n}{s} \geq t, \qquad \frac{n}{s+1} < t,$$

and hence

$$s \leq \frac{n}{t}, \qquad s+1 > \frac{n}{t}, \qquad \text{that is,} \quad s = \left[\frac{n}{t}\right].$$

Now consider the sum

$$\left[\frac{n}{1}\right]\varphi(1) + \left[\frac{n}{2}\right]\varphi(2) + \cdots + \left[\frac{n}{s}\right]\varphi(s) + \cdots + \left[\frac{n}{p}\right]\varphi(p),$$

where p is, of course, assumed to be greater than μ and not greater than n, or rather consider only that part of the sum which extends from the μth term onward:

$$\left[\frac{n}{\mu}\right]\varphi(\mu) + \cdots + \left[\frac{n}{s}\right]\varphi(s) + \cdots + \left[\frac{n}{p}\right]\varphi(p).$$

We may transform this by combining into partial sums those terms for which $[n/s]$ has the same value, and then adding all the partial sums.

For greater uniformity, omit from the first partial sum the term corresponding to $s = \mu$, join it to the already separated first $\mu - 1$ terms, and put

$$\left[\frac{n}{p}\right] = q.$$

Then the partial sums extend respectively from

$s = \mu$	excluded to $s = [n/\nu]$	included, with corresponding value $[n/s] = \nu$,	
$s = [n/\nu]$	to $s = [n/(\nu-1)]$		$\nu - 1$,
$s = [n/(\nu-1)]$	to $s = [n/(\nu-2)]$		$\nu - 2$,
$\cdots$			$\cdots$,
$s = [n/(q+1)]$	to $s = p$		q.

Let $\psi(s)$ denote the summatory term of the function $\varphi(s)$, so that

$$\psi(s) = \sum_{h=1}^{h=s} \varphi(h).$$

The values of the partial sums are then

$$\nu\left(\psi\left[\frac{n}{\nu}\right] - \psi(\mu)\right), \quad (\nu-1)\left(\psi\left[\frac{n}{\nu-1}\right] - \psi\left[\frac{n}{\nu}\right]\right),$$

$$(\nu-2)\left(\psi\left[\frac{n}{\nu-2}\right] - \psi\left[\frac{n}{\nu-1}\right]\right), \quad \dots, \quad q\left(\psi(p) - \psi\left[\frac{n}{q+1}\right]\right),$$

and their combination gives the expression

$$-\nu\psi(\mu) + \psi\left[\frac{n}{\nu}\right] + \psi\left[\frac{n}{\nu-1}\right] + \cdots + \psi\left[\frac{n}{q+1}\right] + q\psi(p).$$

Thus we obtain the general transformation formula

$$\sum_{1}^{p}\left[\frac{n}{s}\right]\varphi(s)=q\psi(p)-\nu\psi(\mu)+\sum_{1}^{\mu}\left[\frac{n}{s}\right]\varphi(s)+\sum_{q+1}^{\nu}\psi\left[\frac{n}{s}\right]. \tag{a}$$

The case that occurs most frequently in the application of this equation is that in which $p=n$ and consequently $q=1$. Under this assumption one obtains

$$\sum_{1}^{n}\left[\frac{n}{s}\right]\varphi(s)=-\nu\psi(\mu)+\sum_{1}^{\mu}\left[\frac{n}{s}\right]\varphi(s)+\sum_{2}^{\nu}\psi\left[\frac{n}{s}\right]. \tag{b}$$

As one sees, the advantage of this transformation is that by it a series whose number of terms is n, and which contains in each term an expression of the form $[n/s]$, is reduced to two others in which the number of terms still containing $[n/s]$ is lowered to order $\sqrt{n}$. A similar transformation can still be carried out if, in the expression $[n/s]$, the denominator s is replaced by a function of s increasing with s. Since such generality is superfluous for our present purpose, we restrict ourselves to the special form of series just considered.

3.

Returning now to the problem treated in Article 1, we obtain, when in equation (b) $\varphi(s)=1$ and hence $\psi(s)=s$, the formula

$$F(n)=\sum_{1}^{n}\left[\frac{n}{s}\right]=-\mu\nu+\sum_{1}^{\mu}\left[\frac{n}{s}\right]+\sum_{1}^{\nu}\left[\frac{n}{s}\right].$$

If in the two sums one puts n/s in place of $[n/s]$, the error is only of order $\sqrt{n}$; and since likewise

$$n\sum_{1}^{\mu}\frac{1}{s}=n\sum_{1}^{\nu}\frac{1}{s}=\frac{1}{2}n\log n+Cn, \qquad \mu\nu=n,$$

it follows, exactly up to an error of order $\sqrt{n}$, that

$$F(n)=n\log n+(2C-1)n.$$

From the asymptotic expression just found for $F(n)$ one can now easily derive an expression for the mean value of the number of divisors. Write the equation, for greater clarity, in the form

$$F(n)=n\log n+(2C-1)n+\xi\sqrt{n},$$

where ξ is unknown but, for however large n may be, remains numerically below a fixed bound that could easily be given. Replace n by $n+s$, with ξ becoming ξ', and divide the difference of the two equations by s. One then obtains for the arithmetic mean of the numbers of factors corresponding to the s numbers $n+1, n+2, \ldots, n+s$:

$$\log(n+s)+2C-1+\frac{n}{s}\log\left(1+\frac{s}{n}\right)+\frac{\xi'\sqrt{n+s}-\xi\sqrt{n}}{s}.$$

If one now lets $s/\sqrt{n}$ increase beyond every bound, the last term tends to zero; that is, the difference between the mentioned arithmetic mean and the expression formed from the remaining terms becomes smaller than any assignable magnitude. Combining the assumption already made with the further one that n/s also exceeds every bound, one obtains, for the mean corresponding to this double condition, the very simple asymptotic value

$$\log n+2C.$$

This also remains valid when n, instead of denoting the number immediately preceding the sequence of s numbers over which the mean is taken, coincides with one of these numbers. For if m is one of them, then $n+t=m$ with $t \le s$, and under the above assumption the difference $\log m - \log n$ tends to zero.

4.

The equation obtained above,

$$\sum_1^n \left[\frac{n}{s}\right] = n\log n + (2C-1)n,$$

in which the error is of order $\sqrt{n}$, gives occasion for an observation at which we shall pause for a moment. Since, on the other hand,

$$\sum_1^n \frac{n}{s} = n\log n + Cn$$

where the error for every n does not exceed a fixed bound, one has, with an error of order $\sqrt{n}$,

$$\sum_1^n \left(\frac{n}{s} - \left[\frac{n}{s}\right]\right) = (1-C)n.$$

Thus the arithmetic mean of the values of the difference

$$\frac{n}{s} - \left[\frac{n}{s}\right],$$

corresponding to $s = 1, 2, \dots, n$, is equal to $1-C$ and hence is less than $\frac{1}{2}$. It is therefore natural to conjecture that, when n is divided successively by the numbers just named, the case in which the remainder lies below half the divisor will occur more often than the opposite case, in which it is equal to or exceeds half the divisor. We shall test the correctness of this conjecture and try to determine the ratio in which the numbers $1, 2, \dots, n$ divide into two groups in this respect.

The number s gives the first or second case according as

$$\frac{n}{s} - \left[\frac{n}{s}\right] < \frac{1}{2} \quad \text{or} \quad \ge \frac{1}{2}.$$

Accordingly one has, respectively,

$$\left[\frac{2n}{s}\right] - 2\left[\frac{n}{s}\right] = 0 \quad \text{or} \quad \left[\frac{2n}{s}\right] - 2\left[\frac{n}{s}\right] = 1.$$

Therefore the number of integers contained in the second group is given by

$$\sum_1^n \left[\frac{2n}{s}\right] - 2\sum_1^n \left[\frac{n}{s}\right],$$

whose second term has already been determined. For the determination of the first one uses equation (a), in which n is replaced by $2n$ and $p = n$, $q = 2$, $\varphi(s) = 1$, $\psi(s) = s$ are put. Thus one obtains

$$\sum_1^n \left[\frac{2n}{s}\right] = 2n - \mu\nu + \sum_1^\mu \left[\frac{2n}{s}\right] + \sum_3^\nu \left[\frac{2n}{s}\right].$$

Since μ is the integer immediately above $\sqrt{2n}$ and $\nu = [2n/\mu]$, $\mu\nu$ differs from $2n$ only by a quantity of order $\sqrt{n}$, and the first two terms cancel. The value of the first sum is, still neglecting terms of order $\sqrt{n}$, given by

$$\sum_1^\mu \left[\frac{2n}{s}\right] = 2n \sum_1^\mu \frac{1}{s} = 2n(\log \mu + C) = n \log 2n + 2Cn.$$

The same value would also hold for the second sum if the two terms corresponding to $s = 1$ and $s = 2$ were not missing. Therefore, to obtain $\sum_1^n [2n/s]$, one must double the expression just found and subtract

$$\left[\frac{2n}{1}\right] + \left[\frac{2n}{2}\right] = 3n.$$

Thus the number of integers in the second group is found to be

$$(\log 4 - 1)n,$$

and for the first group

$$(2 - \log 4)n.$$

For large n, therefore, the number of divisors having the first property is to the number having the second as $2 - \log 4$ is to $\log 4 - 1$.

5.

Now consider the function $f(n)$ which expresses the sum of the divisors of n, or rather, first again as in No. 1, the sum formed from it,

$$F(n) = f(1) + f(2) + \cdots + f(n).$$

By arguments quite similar to those used there, one obtains

$$F(n) = \sum_1^n s \left[\frac{n}{s}\right].$$

Application of equation (b) gives

$$F(n) = -\frac{1}{2}(\mu^2 + \mu)\nu + \sum_1^\mu s \left[\frac{n}{s}\right] + \frac{1}{2} \sum_1^\nu \left(\left[\frac{n}{s}\right]^2 + \left[\frac{n}{s}\right] \right).$$

To see the order of magnitude of the quantities that must be neglected as not completely determinable, first consider the first part of the last sum. Put

$$\left[\frac{n}{s}\right] = \frac{n}{s} - \varepsilon,$$

where ε is a positive proper fraction depending on n and s. Then

$$\sum_1^\nu \left[\frac{n}{s}\right]^2 = \sum_1^\nu \frac{n^2}{s^2} - 2\sum_1^\nu \frac{n}{s}\varepsilon + \sum_1^\nu \varepsilon^2.$$

The second and third terms, whose orders cannot exceed $n \log n$ and $\sqrt{n}$, respectively, must be omitted because of the analytically inexpressible fraction ε occurring in them. The second part of the sum, namely $\sum_1^\nu [n/s]$, which we have already determined and which is of order $n \log n$, is likewise not to be taken into account, although this part can be specified exactly down to and including the first order. Further, neglecting the first order,

$$(\mu^2+\mu)\nu = n^{3/2}, \qquad \sum_1^{\mu} s\left[\frac{n}{s}\right] = n^{3/2};$$

hence, exactly up to an error of order $n\log n$,

$$F(n) = \frac{1}{2}n^{3/2} + \frac{1}{2}n^2\sum_1^{\nu}\frac{1}{s^2}.$$

But by a known formula,

$$\sum_1^{\nu}\frac{1}{s^2} = \frac{\pi^2}{6} - \frac{1}{\nu} + \frac{1}{2\nu^2} - \cdots,$$

and therefore, excluding the first order,

$$\frac{1}{2}n^2\sum_1^{\nu}\frac{1}{s^2} = \frac{\pi^2}{12}n^2 - \frac{1}{2}n^{3/2}.$$

Consequently one finally obtains

$$F(n) = \frac{\pi^2}{12}n^2,$$

where the error does not exceed order $n\log n$.

Using the expression just found to determine the mean value of $f(n)$, one finds for this mean value, namely for

$$\frac{1}{s}\left(f(n+1)+\cdots+f(n+s)\right),$$

the asymptotic expression

$$\frac{\pi^2}{6}n,$$

provided that the simultaneous growth of s and n is understood in such a way that

$$\frac{n}{s} \quad \text{and} \quad \frac{s}{\log n}$$

exceed every finite bound. This result, however, has an essentially different meaning from the one found at the end of No. 3. There the difference between the true mean value and its asymptotic expression became smaller than any given magnitude; here this holds only for the ratio of this difference to the true or approximate mean value.

6.

In the problems treated so far, to which it seems superfluous to add others solved in quite the same way, the function to be approximately determined had directly the form of the series (a). In other cases this function is given by an equation containing such a series, in whose general term the function to be determined occurs; thus one knows only a recurrence relation between successive values of the function. The simplest example of this kind is given by the function $\varphi(n)$, which expresses the number of integers in the sequence $1, 2, \ldots, n$ relatively prime to n and which plays so large a role in number theory. It is well known that this function is expressed by

$$\varphi(n) = n\left(1-\frac{1}{a}\right)\left(1-\frac{1}{b}\right)\left(1-\frac{1}{c}\right)\cdots,$$

where $a, b, c, \dots$ denote the distinct prime numbers dividing n. This formula, however, is not suited to our investigation, and we must choose a different starting point. We begin with the familiar equation

$$\sum \varphi(d) = n,$$

where the summation sign extends over all divisors d of n. Adding this equation and the similar ones for $n-1, n-2, \dots, 1$, the term $\varphi(s)$ on the left, where $s \leq n$,

occurs as many times as there are multiples of s in the sequence $1, 2, \dots, n$, that is, $[n/s]$ times. Thus

$$\sum_1^n \left[\frac{n}{s}\right] \varphi(s) = \frac{1}{2}n^2 + \frac{1}{2}n.$$

Before going further, let us return for a moment to formula (b) and observe that, for some problems, including the one treated in No. 5, it gives a shortcut when the transformation embraces the whole series, although this sacrifices the advantage, indispensable in other cases, of lowering the number of terms to a smaller order. To obtain the modified formula, observe that, if t denotes quite generally one of the numbers $1, 2, 3, \dots, n$, this value is not always found among the terms

$$\left[\frac{n}{1}\right], \quad \left[\frac{n}{2}\right], \quad \dots, \quad \left[\frac{n}{s}\right], \quad \dots, \quad \left[\frac{n}{n}\right],$$

since these terms decrease very rapidly at the beginning of the sequence; nevertheless there is always one and only one term that is not greater than t and is followed by another whose value is smaller than t. As before, the index s of this term must satisfy

$$\frac{n}{s} \geq t, \qquad \frac{n}{s+1} < t,$$

whence

$$s = \left[\frac{n}{t}\right].$$

Thus in general $[n/s] = t$ from $s = [n/(t+1)]$ excluded up to $s = [n/t]$ included; and the sum of all terms in which $[n/s]$ has the value t is

$$\left(\psi\left[\frac{n}{t}\right] - \psi\left[\frac{n}{t+1}\right]\right) t.$$

This expression vanishes and remains correct when no such terms exist, that is, when

$$\left[\frac{n}{t}\right] = \left[\frac{n}{t+1}\right].$$

Combining the values of the expression for $t = 1, 2, \dots, n$, and observing that for $t = n$ the term $\psi\,[n/(t+1)]$ is plainly to be reduced to zero, one obtains

$$\sum_1^n \left[\frac{n}{s}\right] \varphi(s) = \sum_1^n \psi\left[\frac{n}{s}\right]. \tag{c}$$

This transformation is the one we shall now use in our problem. With its help our preceding equation becomes

$$\sum_1^n \psi\left[\frac{n}{s}\right] = \frac{1}{2}n^2 + \frac{1}{2}n.$$

It is soon clear that the asymptotic expression for $\psi(n)$ will have the form αn^2, where α is a constant. One could at first leave this constant undetermined in the following proof, in which case the value $\alpha = 3/\pi^2$ would emerge in the course of the development. Since this value is also easy to foresee, however, we shall for brevity at once consider the expression $\frac{3}{\pi^2}n^2$. Whatever the still unknown function $\psi(n)$ may be, we may set in general

$$\psi(n) = \frac{3}{\pi^2}n^2 + \xi\chi(n),$$

where ξ also depends on n and the function $\chi(n)$ is arbitrary except for the restriction that it always remain positive, increase with its argument, and not vanish at $n = 1$. Suppose $\chi(n)$ has been chosen in the indicated way, and then substitute all values from $n = 1$ to $n = N$ into our equation. The corresponding values of ξ will all be finite. Let A be the greatest numerical value so obtained for ξ. With this assumed, we bring our equation into the form

$$\psi(n) = -\sum_2^n \psi\left[\frac{n}{s}\right] + \frac{1}{2}n^2 + \frac{1}{2}n$$

and consider the values of n lying between $N+1$ and $2N$. Since $[n/s]$ in the sum does not exceed the bound N, the part of our sum arising from substituting the second term of the expression assumed above for $\psi(n)$ is numerically smaller than

$$A\sum_2^n \chi\left(\frac{n}{s}\right).$$

The part arising from the first term, namely

$$\frac{3}{\pi^2}\sum_2^n \left[\frac{n}{s}\right]^2,$$

becomes, if one again writes $[n/s] = n/s - \varepsilon$,

$$\frac{3n^2}{\pi^2}\sum_2^n \frac{1}{s^2} - \frac{6n}{\pi^2}\sum_2^n \frac{\varepsilon}{s} + \frac{3}{\pi^2}\sum_2^n \varepsilon^2.$$

Since

$$\sum_2^n \frac{1}{s^2} = \frac{\pi^2}{6} - 1 + \tau,$$

where τ is of order $1/n$, and since the two other sums cannot exceed the orders $\log n$ and n, respectively, one obtains an equation

$$\psi(n) = \frac{3}{\pi^2}n^2 + \xi,$$

in which ξ is numerically smaller than

$$Pn\log n + A\sum_2^n \chi\left(\frac{n}{s}\right),$$

where P is a sufficiently large constant independent of N. Comparing this result, valid from $n = N+1$ to $n = 2N$, with the expression assumed above for $\psi(n)$,

$$\frac{3}{\pi^2}n^2 + \xi\chi(n),$$

shows that for this new interval the greatest numerical value of ξ does not exceed the maximum, in that interval, of the expression

$$\frac{Pn\log n}{\chi(n)} + \frac{A}{\chi(n)}\sum_2^n \chi\left(\frac{n}{s}\right).$$

If one now gives the previously undetermined function $\chi(n)$ the form of a positive power n^δ, then the expression multiplied by A becomes

$$\sum_2^n \frac{1}{s^\delta} < \sum_2^\infty \frac{1}{s^\delta} = q,$$

and δ can be chosen between 1 and 2 so that the constant q is a proper fraction. On the other hand, N can be chosen so large that for $n \geq N$,

$$\frac{Pn \log n}{n^\delta} < k,$$

where the constant k is arbitrarily small. If A' denotes the greatest numerical value of ξ occurring between $n = N + 1$ and $n = 2N$, then

$$A' < Aq + k.$$

Since k can be made arbitrarily small as N increases, while A does not decrease and q remains constant, for sufficiently large N one has

$$A' < A.$$

That is, the maximum A originally valid up to $n = N$ is also valid up to $2N$, and for the same reason also up to $4N, 8N, \ldots$, and therefore quite generally.

Thus

$$\psi(n) = \frac{3}{\pi^2} n^2$$

with an error that cannot exceed order n^δ, where the constant δ exceeds, however slightly, the value γ given by the equation

$$\sum_2^\infty \frac{1}{s^\gamma} = 1.$$

For the mean value of $\varphi(n)$ this gives the expression

$$\frac{6}{\pi^2} n,$$

whose meaning is clear from the foregoing.

7.

As a final example we choose the function $\varphi(n)$ which denotes the number of all possible decompositions of n into two factors without a common divisor, and which, as is well known, has the expression 2^ρ, where ρ is the number of distinct prime divisors of n. Put again

$$\sum_1^n \varphi(s) = \psi(n).$$

Then $\psi(n)$ stands in a simple relation to the function $F(n)$ treated in Articles 1 and 3. Indeed, $F(n)$ plainly expresses the number of pairs of integers x, y satisfying $xy \leq n$, whereas $\psi(n)$ denotes the number of pairs of relatively prime integers ξ, η for which likewise $\xi\eta \leq n$. Divide the pairs x, y into groups, the sth containing those pairs for which s is the greatest common divisor; thus, if $x = \xi s$, $y = \eta s$, the numbers ξ and η are relatively prime. Dividing the inequality by s^2 gives

$$\xi\eta \leq \frac{n}{s^2} \quad \text{or} \quad \xi\eta \leq \left[\frac{n}{s^2}\right].$$

Conversely, every pair of relatively prime integers ξ, η satisfying this condition gives, by multiplication by s, two integers x, y with greatest common divisor s and with $xy \leq n$. Hence the number of pairs contained in the sth group is expressed by $\psi\left[n/s^2\right]$. Thus one obtains the equation

$$\sum \psi\left[\frac{n}{s^2}\right] = F(n) = n \log n + (2C-1)n,$$

where the sum extends from $s = 1$ to $s = [\sqrt{n}]$, and by the above the neglected term does not exceed order $\sqrt{n}$. It is consequently easy to see that the asymptotic expression for $\psi(n)$ will have the form

$$\alpha n \log n + \beta n,$$

where α and β are two constants still to be determined. If, as in the preceding article, one sets in our equation

$$\psi(n) = \alpha n \log n + \beta n + \xi n^{\delta},$$

and chooses α and β so that the terms of orders $n \log n$ and n cancel, namely

$$\alpha = \frac{6}{\pi^2}, \qquad \beta = \frac{6}{\pi^2}\left(\frac{12C'}{\pi^2} + 2C - 1\right),$$

where C has its former meaning and

$$C' = \sum_{2}^{\infty} \frac{\log s}{s^2},$$

then by arguments entirely analogous to those developed in the preceding article one finds that ξ, however large n may be, remains under a fixed bound, provided only that $\delta > \frac{1}{2}\gamma$, where γ is understood in the above sense. From the expression thus obtained for $\psi(n)$,

$$\alpha n \log n + \beta n,$$

there follows for $\varphi(n)$, in the sense determined by what has just been said, the mean value

$$\frac{6}{\pi^2}\left(\log n + \frac{12C'}{\pi^2} + 2C\right).$$

The question just treated is connected with the mean value given in Article 301 of the *Disquisitiones arithmeticae* for the number of *genera* corresponding to a negative determinant $-n$. According to known theorems, the expression for the number of *genera*, corresponding to the five linear forms

$$n = 8h, \quad n = 8h+4, \quad n = 4h+2, \quad n = 4h+1, \quad n = 4h+3,$$

is respectively

$$\varphi(n), \quad \tfrac{1}{2}\varphi(n), \quad \tfrac{1}{2}\varphi(n), \quad \varphi(n), \quad \tfrac{1}{2}\varphi(n).$$

On the other hand, by obvious modifications of the preceding considerations, whose execution we leave to the reader, one can determine the approximate value of $\psi(n)$ when n has a prescribed linear form, and the sum $\sum \varphi(s) = \psi(n)$ is no longer extended over all the numbers $1, 2, \ldots, n$, but only over the numbers of this form contained in that sequence; from this one can then derive the mean value of $\varphi(n)$. For the linear forms listed above,

$$n = 8h, \quad n = 8h+4, \quad n = 4h+2, \quad n = 4h+1, \quad n = 4h+3,$$

one finds as the mean value of $\varphi(n)$, if for abbreviation one puts

$$\log n + \frac{12C'}{\pi^2} + 2C = \Delta,$$

the following five expressions:

$$\frac{8}{\pi^2}\left(\Delta - \frac{8}{3}\log 2\right), \quad \frac{8}{\pi^2}\left(\Delta - \frac{2}{3}\log 2\right), \quad \frac{8}{\pi^2}\left(\Delta + \frac{1}{3}\log 2\right),$$

$$\frac{4}{\pi^2}\left(\Delta + \frac{4}{3}\log 2\right), \quad \frac{4}{\pi^2}\left(\Delta + \frac{4}{3}\log 2\right).$$

By the foregoing, these expressions, multiplied respectively by $1, \frac{1}{2}, \frac{1}{2}, 1, \frac{1}{2}$, give the mean numbers of the *genera* of determinant $-n$ for the previously enumerated linear forms. Since among every eight consecutive numbers, respectively $1, 1, 2, 2, 2$ are contained in these forms, the sum of our expressions multiplied successively by

$$1 \cdot \tfrac{1}{8}, \quad \tfrac{1}{2} \cdot \tfrac{1}{8}, \quad \tfrac{1}{2} \cdot \tfrac{2}{8}, \quad 1 \cdot \tfrac{2}{8}, \quad \tfrac{1}{2} \cdot \tfrac{2}{8}$$

is, that is,

$$\frac{4}{\pi^2}\left(\Delta - \frac{1}{2}\log 2\right),$$

the desired mean number of *genera* corresponding to the determinant $-n$, when this determinant is thought of generally, that is, no longer restricted to a particular linear form. This agrees with the result given at the cited place. That the same expression also holds for the positive determinant n follows easily from the fact that the mean values of $\varphi(n)$ corresponding to the linear forms $4h + 1$ and $4h + 3$ coincide, and that, on the other hand, the exceptions occurring for square determinants are plainly without influence on the final result.

VII.

On a New Expression for Determining the Density of an Infinitely Thin Spherical Shell, when the Value of Its Potential is Given at Every Point of Its Surface.

By G. Lejeune Dirichlet.

Abhandlungen der Königlich Preussischen Akademie der Wissenschaften, 1850, pp. 99–116.

On a New Expression for Determining the Density of an Infinitely Thin Spherical Shell, when the Value of Its Potential is Given at Every Point of Its Surface.

[Read in the Academy of Sciences on 28 November 1850.[1]]

According to a beautiful theorem first stated and proved by Gauss,[2] every surface can be covered with an infinitely thin mass layer whose potential at each point of the surface assumes an arbitrarily prescribed value, provided that this value varies continuously over the whole extent of the surface. The actual determination of the mass distribution which effects this, however, is at the present state of science possible only for a few special surfaces; among these, as Gauss already observed, is the whole spherical surface.

Let R denote the radius of the sphere, r the distance of any point in space from the centre, θ the angle between r and a fixed straight line, and φ the angle in the surface between the plane determined by this line and r and a fixed plane. The potential value V, prescribed on the entire surface and expressed by θ and φ, may then be expanded in the known spherical functions. Let

$$V = \sum X_n$$

be this expansion, where the summation sign extends from $n = 0$ to $n = \infty$, and let

$$\rho = \sum Y_n$$

be the analogous expansion of the density ρ to be determined. With the help of the latter, the potential v at any point in space can be represented.[3] Of the two expressions for it, one valid in the interior and the other in the exterior space, either would suffice for our purpose; the former is

$$v = 4\pi R \sum \frac{1}{2n+1} \left(\frac{r}{R}\right)^n Y_n.$$

Since this expression remains valid up to $r = R$, in which case v becomes V, and since the same function is capable of only one expansion in spherical functions, comparison with the first series

[1]The Academy report for 1850, p. 464, states: "Mr. Dirichlet gave a lecture on a new expression for the mass distribution on a spherical surface when the potential at every point of the surface is to receive an arbitrarily prescribed value." K.

[2]General theorems concerning the attractive and repulsive forces acting in inverse ratio to the square of the distance, by C. F. Gauss.

[3]Mécanique céleste, Book III, No. 13.

gives

$$Y_n = \frac{2n+1}{4\pi R} X_n,$$

and hence

$$\rho = \frac{1}{4\pi R} \sum (2n+1) X_n.$$

In an earlier paper[4] it was shown that every function prescribed arbitrarily over the whole spherical surface, that is, from $\theta = 0$, $\varphi = 0$ to $\theta = \pi$, $\varphi = 2\pi$, can be developed convergently in spherical functions, provided only that it nowhere becomes infinite. The convergence of the series for V, which is the starting point of the solution just described, is therefore beyond doubt. The matter is different for the series $\sum Y_n$ assumed for the density ρ and then determined by comparison with it. Since the existence of a completely determined mass distribution corresponding to the prescribed potential value V is compatible with the density becoming infinite at particular points or along particular lines, it remains entirely uncertain in all such cases whether the series preserves its convergence at all places where the density remains finite and whether it actually represents the density. It seemed to me not uninteresting to submit this question to closer examination, for which my earlier paper supplied the necessary aids, especially since this investigation allowed one to expect a simplification of the expression just found for ρ. That expression involves a fourfold infinite operation: a double summation and likewise a double integration. That the latter cannot be removed from the expression is clear, since the density at each point must depend on all potential values prescribed, up to continuity, over the whole extent of the surface. On the other hand, it has turned out that the density can always be represented without series by a double integral, which in many cases can even be reduced to a single integral.

1.

Put

$$V = f(\theta, \varphi).$$

It is known that for the general term X_n one has

$$X_n = \frac{2n+1}{4\pi} \iint f(\theta', \varphi') P_n(\cos\omega) \sin\theta' \, d\theta' \, d\varphi',$$

where

$$\cos\omega = \cos\theta\cos\theta' + \sin\theta\sin\theta'\cos(\varphi - \varphi')$$

and where $P_n(\cos\omega)$ denotes the coefficient of α^n in the expansion of the radical expression

$$\frac{1}{\sqrt{1 - 2\alpha\cos\omega + \alpha^2}}.$$

The double integration extends from

$$\theta' = 0,\ \varphi' = 0 \qquad \text{to} \qquad \theta' = \pi,\ \varphi' = 2\pi.$$

If the divisor R is omitted, or equivalently if the radius of the sphere is set equal to unity, the general term of the series representing ρ becomes

$$\frac{(2n+1)^2}{(4\pi)^2} \iint f(\theta', \varphi') P_n(\cos\omega) \sin\theta' \, d\theta' \, d\varphi'.$$

[4]Sur les séries dont le terme général dépend de deux angles, et qui servent à exprimer des fonctions arbitraires entre des limites données. Crelle's Journal, vol. XVII. Vol. I, p. 283 of this edition of G. Lejeune Dirichlet's Werke. K.

It will plainly be enough to examine this series for the case $\theta = 0$, that is, for the pole p of the spherical polar coordinates θ, φ, since, as in the earlier paper, the result found for the point p may be transferred immediately to any other point m of the surface. Then

$$\cos\omega = \cos\theta',$$

and $P_n(\cos\omega)$ is independent of φ'. Put

$$\frac{1}{2\pi}\int_0^{2\pi} f(\theta', \varphi')\, d\varphi' = F(\theta').$$

Thus $F(\theta')$ denotes the mean value of the potential V on the circle described about p as centre with spherical radius θ'. Writing γ instead of θ', in agreement with the notation of the earlier paper to which we shall often refer, the general term becomes

$$\frac{(2n+1)^2}{8\pi}\int_0^{\pi} F(\gamma) P_n(\cos\gamma)\sin\gamma\, d\gamma.$$

If $(2n+1)^2$ is decomposed into its components $4n^2$, $4n$, 1, then the general term splits into three others, and consequently the series itself into three partial series. Since the convergence of the second and third of these series has already been shown in the earlier paper, we have only to consider the first, whose general term is

$$\frac{1}{2\pi} n^2 \int_0^{\pi} F(\gamma) P_n(\cos\gamma)\sin\gamma\, d\gamma.$$

Taking the sum of the first $n+1$ terms, expressing $P_n(\cos\gamma)$ by means of a definite integral according to equation (3) of the earlier paper,[5] and reversing the order of the two integrations, one obtains

$$\frac{1}{\pi^2}\int_0^{\pi} (\cos\psi + 4\cos 2\psi + \cdots + n^2\cos n\psi)\Pi(\psi)\, d\psi,$$

where, as before,

$$\Pi(\psi) = \sin\frac{\psi}{2}\int_0^{\psi} F(\gamma)\sin\gamma\frac{d\gamma}{\Delta} + \cos\frac{\psi}{2}\int_{\psi}^{\pi} F(\gamma)\sin\gamma\frac{d\gamma}{E},$$

$$\Delta = \sqrt{2(\cos\gamma - \cos\psi)}, \qquad E = \sqrt{2(\cos\psi - \cos\gamma)}.$$

Since the integrals contained in $\Pi(\psi)$ are functions of ψ which remain continuous from $\psi = 0$ to $\psi = \pi$ (by §3 of the earlier paper), the same property belongs to the function $\Pi(\psi)$ itself.

The present investigation also requires discussion of the derivative $\Pi'(\psi)$. In forming it, because of the denominators Δ and E contained in the integrals, a preliminary transformation by integration by parts is necessary. Taking into account that $F(\gamma)$, from its origin as the continuous potential value $f(\theta, \varphi)$, is itself continuous, this double operation gives

$$\Pi'(\psi) = (F(0) - F(\pi))\sin\psi + \frac{1}{2}\cos\frac{\psi}{2}\int_0^{\psi} F'(\gamma)\Delta\, d\gamma + \frac{1}{2}\sin\frac{\psi}{2}\int_{\psi}^{\pi} F'(\gamma) E\, d\gamma$$
$$+ \sin\frac{\psi}{2}\sin\psi\int_0^{\psi} F'(\gamma)\frac{d\gamma}{\Delta} + \cos\frac{\psi}{2}\sin\psi\int_{\psi}^{\pi} F'(\gamma)\frac{d\gamma}{E}.$$

We now assume that $F'(\gamma)$ remains finite everywhere from $\gamma = 0$ to $\gamma = \pi$. Then all components of $\Pi'(\psi)$, and hence $\Pi'(\psi)$ itself, will be finite and continuous from $\psi = 0$ to $\psi = \pi$. For the first three terms this is evident; for the last two, arguments very similar to those cited above show that the integrals contained in them retain this double property so long as, in the first, $\psi < \pi$

[5] Vol. I, p. 291 of this edition of G. Lejeune Dirichlet's Werke. K.

and, in the second, $\psi > 0$ is assumed. For $\psi = \pi$ and $\psi = 0$ these integrals may respectively take infinite values, but, as is easy to see, their order cannot exceed

$$\log\left(\frac{1}{\cos\frac{\psi}{2}}\right) \qquad \text{and} \qquad \log\left(\frac{1}{\sin\frac{\psi}{2}}\right),$$

so that the terms themselves become zero.

For what follows one must also know the second derivative $\Pi''(\psi)$ for the special value $\psi = 0$. Since $\Pi'(0) = 0$, the desired value is most easily found by letting ψ become infinitely small in the quotient $\frac{1}{\psi}\Pi'(\psi)$. Thus one obtains

$$\Pi''(0) = F(0) - F(\pi) + \frac{1}{2}\int_0^\pi F'(\gamma)\sin\frac{\gamma}{2}\,d\gamma + \frac{1}{2}\int_0^\pi \frac{F'(\gamma)}{\sin\frac{\gamma}{2}}\,d\gamma,$$

or, if the third term is integrated by parts,

$$\Pi''(0) = F(0) - \frac{1}{2}F(\pi) - \frac{1}{4}\int_0^\pi F(\gamma)\cos\frac{\gamma}{2}\,d\gamma + \frac{1}{2}\int_0^\pi \frac{F'(\gamma)}{\sin\frac{\gamma}{2}}\,d\gamma.$$

If, in addition to the finiteness of $F'(\gamma)$ just assumed and the consequent continuity of $\Pi'(\psi)$, a definite finite value also exists for $\Pi''(0)$, or what is the same, if the second integral has such a value, then our series will always converge and its sum will be easy to state. Indeed, for shortness put

$$X(\psi) = \frac{\sin(2n+1)\frac{\psi}{2}}{2\sin\frac{\psi}{2}} = \frac{1}{2} + \cos\psi + cos 2\psi + \cdots + \cos n\psi.$$

The above expression for the sum of the first $n+1$ terms is then

$$-\frac{1}{\pi^2}\int_0^\pi \Pi(\psi)X''(\psi)\,d\psi.$$

Since the boundary term emerging after twice integrating by parts,

$$\Pi(\psi)X'(\psi) - \Pi'(\psi)X(\psi),$$

is continuous and vanishes at both limits, this operation gives

$$-\frac{1}{\pi^2}\int_0^\pi \Pi''(\psi)\frac{\sin(2n+1)\frac{\psi}{2}}{2\sin\frac{\psi}{2}}\,d\psi.$$

By a known theorem, this expression tends, as n increases, even when $\Pi''(\psi)$ becomes infinite for one or more non-zero values of ψ, to the limit

$$\begin{aligned}-\frac{1}{2\pi}\Pi''(0) = &-\frac{1}{2\pi}F(0) + \frac{1}{4\pi}F(\pi) + \frac{1}{8\pi}\int_0^\pi F(\gamma)\cos\frac{\gamma}{2}\,d\gamma \\ &- \frac{1}{4\pi}\int_0^\pi \frac{F'(\gamma)}{\sin\frac{\gamma}{2}}\,d\gamma.\end{aligned}$$

Adding to this value those of the two other series, as found in the earlier paper, gives for the density at p

$$\rho = \frac{1}{4\pi}\left[F(\pi) - \int_0^\pi \frac{F'(\gamma)}{\sin\frac{\gamma}{2}}\,d\gamma\right].$$

It remains to examine how the series expressing the density behaves, as regards convergence, in the case where the integral

$$\int_0^\pi \frac{F'(\gamma)}{\sin\frac{\gamma}{2}}\,d\gamma$$

and therefore also the expression just found for ρ retains a definite finite value, while the function $F'(\gamma)$ becomes infinite for one or more non-zero values of γ. According to the above, the condition for convergence then consists solely in this: the function $\Pi''(\psi)$ derived from $F'(\gamma)$ must remain finite and continuous from $\psi = 0$ to $\psi = \pi$. A more exact consideration of the way $\Pi''(\psi)$ is formed shows that, if the becoming infinite of $F'(\gamma)$ occurs in such a way that for every value c at which $F'(\gamma)$ takes an infinitely large value, $\sqrt{\varepsilon}F'(c \pm \varepsilon)$ is itself infinitely small for infinitely small ε, then the continuity of $\Pi''(\psi)$ obtains just as in the previously examined case of an everywhere finite function $F'(\gamma)$, and with it the convergence of the series representing the density. In contrast, in general divergence occurs when the condition just stated is no longer fulfilled. Although the proof of this result presents no essential difficulty, it would require too much space and would be of too little interest to carry it out here. For the secure use of the series it is enough to know, from the preceding discussion, the only cases in which the convergence of the series can cease. Moreover, an occasion will arise below to demonstrate the divergence actually occurring in such a case by means of a very simple example.

2.

Since the preceding derivation of the finite expression found for ρ loses its validity when the series ceases to converge, a proof of this expression applicable to all cases is still needed.

According to a known theorem, the derivative $\frac{\partial v}{\partial r}$, when θ and φ are kept constant and r is allowed to approach unity, tends to two different limits according as $r < 1$ or $r > 1$ is assumed throughout. If these limiting values are denoted respectively by K and L, then the density at the corresponding point of the surface is determined by

$$\rho = \frac{1}{4\pi}(K - L).$$

Between the limits K, L and the potential value V on the surface there is a simple relation, so that only one of the two limiting values has to be found. Namely, if v and v_1 are the potential values for two points on the same radius vector, their distances from the centre, r and r_1, satisfying

$$rr_1 = 1,$$

then, as is readily seen,

$$v_1 = \frac{1}{r_1}v,$$

and therefore

$$\frac{\partial v_1}{\partial r_1} = -\frac{1}{r_1^2}v + \frac{1}{r_1}\frac{\partial v}{\partial r}\frac{\partial r}{\partial r_1} = -\frac{1}{r_1^2}v - \frac{1}{r_1^3}\frac{\partial v}{\partial r}.$$

Thus one sees that

$$L = -V - K$$

and

$$\rho = \frac{1}{4\pi}(V + 2K).$$

To determine K it suffices to use the expression for v valid in the interior space:

$$v = \frac{1 - r^2}{4\pi}\iint \frac{f(\theta', \varphi')\sin\theta'\, d\theta'\, d\varphi'}{(1 - 2r\cos\omega + r^2)^{3/2}}.$$

This is easily proved without developing in series. It is enough to observe that this expression, as r approaches its upper limit 1, passes by a known and often treated theorem into $f(\theta, \varphi) = V$, and that on the other hand it satisfies the partial differential equation first set up by Laplace for v, since

$$\frac{1-r^2}{(1-2r\cos\omega+r^2)^{3/2}},$$

considered as a function of the polar coordinates r, θ, φ, is a particular integral of this equation. In the case $\theta = 0$, the expression takes, writing γ again instead of θ' and keeping the earlier notation, the simple form

$$v = \frac{1}{2}(1-r^2)\int_0^\pi \frac{F(\gamma)\sin\gamma\,d\gamma}{(1-2r\cos\gamma+r^2)^{3/2}}.$$

If one differentiated the expression in this form, the limiting value of $\frac{\partial v}{\partial r}$ would be difficult to determine. It is therefore expedient first to perform an integration by parts. One obtains

$$2v = \left(\frac{1}{r}+1\right)F(0) - \left(\frac{1}{r}-1\right)F(\pi) + \left(\frac{1}{r}-r\right)\int_0^\pi \frac{F'(\gamma)\,d\gamma}{(1-2r\cos\gamma+r^2)^{1/2}},$$

where passage to the limit after differentiation now presents no further difficulty and gives

$$K = -\frac{1}{2}F(0) + \frac{1}{2}F(\pi) - \frac{1}{2}\int_0^\pi \frac{F'(\gamma)}{\sin\frac{\gamma}{2}}\,d\gamma.$$

Taking into account that for the point p,

$$V = F(0),$$

it follows that

$$\rho = \frac{1}{4\pi}\left[F(\pi) - \int_0^\pi \frac{F'(\gamma)}{\sin\frac{\gamma}{2}}\,d\gamma\right],$$

which agrees with the formula derived from the series.

3.

Transferring the result found for the point corresponding to $\theta = 0$ to an arbitrary point m, one obtains the following general rule for determining the density.

By a first integration, find for the circle described on the surface about m as centre with arbitrary spherical radius λ the mean value of the given potential V. Denote this mean by $\varphi(\lambda)$. Then the density sought is given by

$$\rho = \frac{1}{4\pi}\left[\varphi(\pi) - \int_0^\pi \frac{\varphi'(\lambda)}{\sin\frac{\lambda}{2}}\,d\lambda\right].$$

One may also remark that, by the meaning of the function $\varphi(\lambda)$, the special value $\varphi(\pi)$ denotes the potential value at the point antipodal to m. It hardly needs saying that the function $\varphi(\lambda)$ will be different for every point m, or, in other words, that besides λ it will also contain the two quantities which determine the position of m on the surface.

It is perhaps not superfluous, however, to forestall an error which can easily arise from an inattentive consideration of the result just stated. At first sight, because the denominator under the integral sign vanishes at the lower limit, one is tempted to suppose that the integral remains finite only for special positions of the point m, but in general becomes infinite; in fact, precisely the reverse is the case. First, by the continuity of $\varphi(\lambda)$ it is easy to see that the part of the integral extending from any positive value δ, however small, to the upper limit π always remains determinate and finite, however often $\varphi'(\lambda)$ itself may become infinite in this interval. That the same is also true, apart from singular cases, for the part extending from 0 to δ has its reason in the fact that for small values of λ, $\varphi'(\lambda)$ is generally of the same first order as the denominator. If the potential value V is represented as a function $\chi(\lambda, \psi)$ of spherical polar coordinates λ, ψ whose pole is the point m, so that

$$\varphi(\lambda) = \frac{1}{2\pi} \int_0^{2\pi} \chi(\lambda, \psi)\, d\psi,$$

then the property just mentioned does not appear only because $\chi(\lambda, \psi)$ is not arbitrary, but must satisfy the special conditions of being periodic with respect to ψ and becoming independent of ψ for $\lambda = 0$. One is immediately convinced of the assertion made above if one projects the spherical surface onto an arbitrary plane, whose points are referred to two rectangular axes t, u, and then expresses V at each point by the coordinates t, u of its projection. This representation has the disadvantage that, in order to cover the whole surface, two functions of t and u must be considered; but this disadvantage disappears for our purpose, which requires consideration only of an arbitrarily small surface element containing the point m. Taking the tangent plane at m as the plane of projection, m as the origin of coordinates, and putting $V = \chi(t, u)$, this new function has to satisfy no condition other than continuity. Since evidently one may take

$$t = \sin\lambda \cos\psi, \qquad u = \sin\lambda \sin\psi,$$

the function $\varphi(\lambda)$ assumes the form

$$\varphi(\lambda) = \frac{1}{2\pi} \int_0^{2\pi} \chi(\sin\lambda\cos\psi, \sin\lambda\sin\psi)\, d\psi.$$

In this form the occurrence of the above property is evident, provided the partial derivatives of $\chi(t, u)$ up to and including the second order, for the case where $t = 0$ and $u = 0$ simultaneously, retain definite finite values. Moreover, for the finiteness of ρ it is not even necessary that $\varphi'(\lambda)$ be of the first order for small values of the variable; it is enough that the order of $\varphi'(\lambda)$ agree with that of some positive power of λ.

As with the becoming infinite of ρ and the consequent divergence of the series, so also with the case where the series ceases to converge while the density remains finite. This case is even more restricted than the one just discussed, as is easily seen by closer consideration of the conditions on which it depends.

4.

In order to illustrate the actual occurrence of divergence of the series for ρ in the exceptional cases indicated above by a simple example, suppose that the potential value V does not contain the angle φ and is represented by $f(\theta)$. Then, as is known, the spherical-function series take a simpler form, and one has

$$V = \frac{1}{2}\sum(2n+1)A_n P_n(\cos\theta), \qquad \rho = \frac{1}{8\pi}\sum(2n+1)^2 A_n P_n(\cos\theta),$$

where the general coefficient A_n is given by

$$A_n = \int_0^{\pi} f(\theta) P_n(\cos\theta)\sin\theta\, d\theta.$$

Put

$$f(\theta) = \sqrt{\cos\theta} \quad \text{as long as } 0 < \theta < \frac{1}{2}\pi, \qquad f(\theta) = 0 \quad \text{when } \theta \text{ lies between } \frac{1}{2}\pi \text{ and } \pi.$$

Then the requirement of continuity is satisfied, and the integral

$$A_n = \int_0^{\frac{1}{2}\pi} P_n(\cos\theta)\sqrt{\cos\theta}\sin\theta\,d\theta$$

has to be determined. Since its evaluation by means of the known expressions for $P_n(\cos\theta)$ does not immediately give the result in the simplest form, the following way is preferable.

From

$$\sum P_n(\cos\theta)\alpha^n = \frac{1}{\sqrt{1-2\alpha\cos\theta+\alpha^2}},$$

the required integral is the coefficient of α^n in the expansion of

$$\int_0^{\frac{1}{2}\pi} \frac{\sqrt{\cos\theta}\sin\theta\,d\theta}{\sqrt{1-2\alpha\cos\theta+\alpha^2}},$$

where the proper fraction α may be regarded as positive. By the substitution $t=\sqrt{\cos\theta}$ and by carrying out the integration, one obtains

$$2\int_0^1 \frac{t^2\,dt}{\sqrt{1-2\alpha t^2+\alpha^2}} = \sqrt{\frac{1}{8\alpha^3}}\left((1+\alpha^2)\arcsin\sqrt{\frac{2\alpha}{1+\alpha^2}} - (1-\alpha)\sqrt{2\alpha}\right).$$

For convenience in developing the expression, put

$$z = \arcsin\sqrt{\frac{2\alpha}{1+\alpha^2}}.$$

Then differentiating gives

$$\frac{dz}{d\alpha} = \frac{1+\alpha}{1+\alpha^2}\sqrt{\frac{1}{2\alpha}} = \frac{1}{\sqrt{2}}(\alpha^{-\frac{1}{2}}+\alpha^{\frac{1}{2}}-\alpha^{\frac{3}{2}}-\alpha^{\frac{5}{2}}+\cdots),$$

and, on integrating and observing that z vanishes with α,

$$\arcsin\sqrt{\frac{2\alpha}{1+\alpha^2}} = \sqrt{2}\left(\alpha^{\frac{1}{2}}+\frac{1}{3}\alpha^{\frac{3}{2}}-\frac{1}{5}\alpha^{\frac{5}{2}}-\frac{1}{7}\alpha^{\frac{7}{2}}+\cdots\right).$$

Substitution gives for the required expansion

$$-\frac{1}{2}\left(\left(\frac{1}{-1}-\frac{1}{3}\right)-\left(\frac{1}{1}-\frac{1}{5}\right)\alpha-\left(\frac{1}{3}-\frac{1}{7}\right)\alpha^2+\left(\frac{1}{5}-\frac{1}{9}\right)\alpha^3+\cdots\right),$$

where the signs inside the parentheses form the four-term period

$$+\ -\ -\ +,$$

which may be represented by $(-1)^{\frac{1}{2}n(n+1)}$. Thus

$$A_n = -(-1)^{\frac{1}{2}n(n+1)}\frac{2}{(2n-1)(2n+3)},$$

and the preceding series take the form

$$V = -\sum(-1)^{\frac{1}{2}n(n+1)}\frac{2n+1}{(2n-1)(2n+3)}P_n(\cos\theta),$$

$$\rho = -\frac{1}{4\pi}\sum(-1)^{\frac{1}{2}n(n+1)}\frac{(2n+1)^2}{(2n-1)(2n+3)}P_n(\cos\theta).$$

From the remarks in the preceding article, one sees without difficulty in the present case that the derivative $\varphi'(\lambda)$ becomes infinite in such a way that the condition stated at the end of Article 1 is not fulfilled only when $\lambda = \frac{1}{2}\pi$ and the point m corresponds to $\theta = 0$ or $\theta = \pi$; in these cases, however, $\varphi'(\lambda)$ is of order λ for small values of λ, and ρ therefore retains a definite finite value. One also sees that the density becomes infinite only at the equator, or for $\theta = \frac{1}{2}\pi$, since then $\varphi'(\lambda)$ is of order λ^{-1} for small λ. In these three cases the series therefore ceases to converge. This is immediately verified if one takes into account that $P_n(\cos\theta)$ has the values 1 and $(-1)^n$ respectively for $\theta = 0$ and $\theta = \pi$, while for $\theta = \frac{1}{2}\pi$ it vanishes when n is odd and, when n is even, equals

$$\frac{1\cdot 3\cdot 5\cdots(n-1)}{2\cdot 4\cdot 6\cdots n}(-1)^{\frac{n}{2}},$$

which, as is known, is of order $n^{-\frac{1}{2}}$ for large n.

In the first two cases, where ρ retains a definite finite value, the series has the character of an oscillating one: among every four consecutive terms, two have positive and two negative sign, and their terms tend to unity as a limit. In the third case, all terms have the same sign and yield an infinitely large sum.

5.

Although the method by which we have just determined the coefficient A_n is no longer applicable if, while retaining the assumption $f(\theta) = 0$ for the second part of the interval, one takes $f(\theta) = \cos^k\theta$ in the first part, where k denotes an arbitrary positive constant, the very simple form of the result found for the special value $k = \frac{1}{2}$ leads one to expect something similar in the more general case. Since the simplest expression for A_n is somewhat hidden, we pause briefly to find it. Poisson, who in one of his papers[6] chose precisely the function just defined as an example of development in spherical functions, gives the coefficient A_n the form

$$A_n = \frac{1\cdot 3\cdots(2n-1)}{1\cdot 2\cdots n}\left(\frac{1}{k+n+1} - \frac{n(n-1)}{2(2n-1)}\frac{1}{k+n-1} + \frac{n(n-1)(n-2)(n-3)}{2\cdot 4(2n-1)(2n-3)}\frac{1}{k+n-3} - \cdots\right).$$

This is the form in which the coefficient is obtained directly when the usual expression for $P_n(\cos\theta)$ is used in the expansion. It does not reveal the considerable simplification of which it is capable, and that simplification would not easily be suspected unless one were led to it by a special case such as the one above. The following very short, though not elementary, route leads to the simplest expression for A_n.

By §1 of the earlier paper,[7] the two integrals

$$\frac{2}{\pi}\int_0^\gamma \frac{\cos n\psi\cos\frac{\psi}{2}}{\sqrt{2(\cos\psi-\cos\gamma)}}\,d\psi, \qquad -\frac{2}{\pi}\int_0^\gamma \frac{\sin n\psi\sin\frac{\psi}{2}}{\sqrt{2(\cos\psi-\cos\gamma)}}\,d\psi$$

are respectively the coefficients of α^n in the expansions of

$$\frac{1}{2}+\frac{1}{2}\frac{1+\alpha}{\sqrt{1-2\alpha\cos\gamma+\alpha^2}}, \qquad -\frac{1}{2}+\frac{1}{2}\frac{1-\alpha}{\sqrt{1-2\alpha\cos\gamma+\alpha^2}}.$$

Adding gives

$$P_n(\cos\gamma) = \frac{2}{\pi}\int_0^\gamma \frac{\cos(n+\frac{1}{2})\psi}{\sqrt{2(\cos\psi-\cos\gamma)}}\,d\psi.$$

[6] Connaissance des temps pour l'an 1829.

[7] Vol. I, pp. 292–294 of this edition of G. Lejeune Dirichlet's Werke. K.

Multiplying this equation by $\cos^k \gamma \sin \gamma \, d\gamma$, then integrating from $\gamma = 0$ to $\gamma = \frac{1}{2}\pi$ and reversing the order of integrations on the right, yields

$$A_n = \frac{2}{\pi} \int_0^{\frac{1}{2}\pi} d\psi \, \cos(n + \tfrac{1}{2})\psi \int_\psi^{\frac{1}{2}\pi} \frac{\cos^k \gamma \sin \gamma}{\sqrt{2(\cos\psi - \cos\gamma)}} \, d\gamma.$$

If one introduces a new variable t by $\cos\gamma = t\cos\psi$, the two integrations separate:

$$A_n = \frac{\sqrt{2}}{\pi} \int_0^1 \frac{t^k}{\sqrt{1-t}} \, dt \cdot \int_0^{\frac{1}{2}\pi} \cos^{k+\frac{1}{2}} \psi \cos(n + \tfrac{1}{2})\psi \, d\psi.$$

Now, by a known formula,

$$\int_0^{\frac{1}{2}\pi} \cos^{p-1} \psi \cos q\psi \, d\psi = \frac{\Gamma(p)}{\Gamma\left(\frac{1}{2}(1+p+q)\right)\Gamma\left(\frac{1}{2}(1+p-q)\right)} \cdot \frac{\pi}{2^p},$$

where the constant p must be positive, and the transcendental function $\Gamma(l)$, when the argument l becomes negative, is to be understood by the relation $\Gamma(l+1) = l\Gamma(l)$, or, what is the same thing, by Gauss's definition, which comprises positive and negative arguments uniformly. By this formula and by the known representation of a Eulerian integral of the first kind by three of the second, our equation becomes

$$A_n = \frac{\Gamma(k+1)}{\Gamma\left(\frac{1}{2}(k+n+3)\right)\Gamma\left(\frac{1}{2}(k-n+2)\right)} \cdot \frac{\sqrt{\pi}}{2^{k+1}},$$

or

$$A_n = \frac{k(k-1)(k-2)\cdots(k-n+1)}{\frac{1}{2}(k+n+1)\left(\frac{1}{2}(k+n+1)-1\right)\cdots\left(\frac{1}{2}(k+n+1)-n\right)} \cdot \frac{\Gamma(k-n+1)}{\Gamma\left(\frac{1}{2}(k-n+1)\right)\Gamma\left(\frac{1}{2}(k-n+2)\right)} \cdot \frac{\sqrt{\pi}}{2^{k+1}}.$$

If for the second factor one substitutes its known value $2^{k-n}\pi^{-\frac{1}{2}}$, this gives

$$A_n = \frac{k(k-1)(k-2)\cdots(k-n+1)}{(k+n+1)(k+n-1)(k+n-3)\cdots(k-n+1)},$$

where the factors in the denominator decrease by two units.

It hardly needs mention that this simple expression can be verified with the greatest ease. For instance, it may be decomposed into partial fractions in order to recognize its agreement with the expression cited above. If one does not wish to use the latter for this verification, one may use instead the equation

$$(n+1)A_{n+1,k} = (2n+1)A_{n,k+1} - nA_{n-1,k},$$

which follows immediately from the relation between every three consecutive coefficients in the expansion of $P_n(\cos\gamma)$.

To simplify the expression still further, one distinguishes whether n is even or odd and obtains respectively

$$A_n = \frac{k(k-2)\cdots(k-n+2)}{(k+1)(k+3)\cdots(k+n+1)}, \qquad A_n = \frac{(k-1)(k-3)\cdots(k-n+2)}{(k+2)(k+4)\cdots(k+n+1)}.$$

With the help of these expressions and the known formulae which give the approximate values of factorials containing a very large number of factors, one may easily verify that the special case $k = \frac{1}{2}$ examined above is exactly the limiting case for the divergence of the series occurring at $\theta = 0$ and $\theta = \pi$, and that the divergence ceases as soon as $k > \frac{1}{2}$. The situation is different for the value $\theta = \frac{1}{2}\pi$: at that place the divergence and the infinitely large value of the density persist as long as $k > 1$ is not assumed.

6.

We shall now make one application of the finite expression found for the density to a special case. Let V again be regarded as a mere function of θ, which may be put in the form $f(\cos\theta)$. Then ρ too depends only on the spherical distance θ of the point m from the pole p. On the circle described about m as centre with spherical radius λ, the fundamental formula of trigonometry gives

$$V = f(\cos\theta\cos\lambda + \sin\theta\sin\lambda\cos\psi),$$

where ψ denotes the angle between an arbitrary radius λ and the one directed from m to p. Since this expression has the same value for ψ and $-\psi$, the integral expressing the mean value $\varphi(\lambda)$ may be taken from $\psi = 0$ to $\psi = \pi$ and then doubled. Thus

$$\varphi(\lambda) = \frac{1}{\pi}\int_0^\pi f(\cos\theta\cos\lambda + \sin\theta\sin\lambda\cos\psi)\,d\psi.$$

Whenever $\varphi'(\lambda)$ can be represented without an integral sign, which is especially the case when $f(z)$, or more generally $f'(z)$, is a rational function of z, whether the form of that function remains the same throughout the whole interval between $z = -1$ and $z = +1$ or changes in places without violating continuity, ρ can be reduced to a single quadrature or even expressed without any integral sign.[8]

Among the cases belonging here we consider only the one in which

$$f(\cos\theta) = \cos\theta$$

as long as $\theta < \frac{1}{2}\pi$, and

$$f(\cos\theta) = 0$$

when $\theta > \frac{1}{2}\pi$. The integral then extends only over that part of the interval between $\psi = 0$ and $\psi = \pi$ in which

$$\cos\theta\cos\lambda + \sin\theta\sin\lambda\cos\psi$$

assumes positive values. Suppose first that $\theta < \frac{1}{2}\pi$; the case $\theta > \frac{1}{2}\pi$ is easily reduced to this. A simple discussion of the preceding expression shows that, so long as λ is less than $\frac{1}{2}\pi - \theta$, the integral is to be taken from $\psi = 0$ to $\psi = \pi$; for values of λ between $\frac{1}{2}\pi - \theta$ and $\frac{1}{2}\pi + \theta$, the limits of the integral are 0 and the angle ψ_1 lying between 0 and π determined by

$$\cos\theta\cos\lambda + \sin\theta\sin\lambda\cos\psi_1 = 0;$$

finally, for $\lambda > \frac{1}{2}\pi + \theta$, the integral vanishes, since the preceding expression is then always negative. Thus, in the three intervals just distinguished, the function $\varphi(\lambda)$ is

$$\cos\theta\cos\lambda, \qquad \frac{1}{\pi}\int_0^{\psi_1}(\cos\theta\cos\lambda + \sin\theta\sin\lambda\cos\psi)\,d\psi, \qquad 0.$$

In the second case the actual integration is postponed until after differentiation, in order to shorten the calculation. On differentiating, the term arising from the variability of the upper limit ψ_1 vanishes by the equation determining ψ_1, and one obtains the following three expressions for $\varphi'(\lambda)$:

$$-\cos\theta\sin\lambda, \qquad \frac{1}{\pi}(-\psi_1\cos\theta\sin\lambda + \sin\psi_1\sin\theta\cos\lambda), \qquad 0.$$

[8] In the general case, where V contains both angles θ, φ, the reduction of ρ to a single quadrature always takes place if, putting $x = \cos\theta$, $y = \sin\theta\cos\varphi$, $z = \sin\theta\sin\varphi$, one can represent V by a function $f(x, y, z)$ of the quantities x, y, z whose three first partial derivatives are rational, integral or fractional functions of x, y, z; again the form of $f(x, y, z)$ may be different in different parts of the surface. This is a result remarkable for its great range.

These formulas also show in passing that, at least so long as m remains on the hemisphere for which they apply, $\varphi'(\lambda)$ does not become infinite and, for a small λ for which the first formula holds, remains of the first order. In this respect there is an exception only for $\theta = \frac{1}{2}\pi$: then the two outer intervals disappear and $\varphi'(\lambda)$ everywhere has the value $\frac{1}{\pi}\cos\lambda$, which for small values of λ is of order zero; hence an infinitely large density occurs at the equator.

According to the expressions found for $\varphi'(\lambda)$, the integral contained in the general expression for ρ splits into two others, extending respectively from 0 to $\frac{1}{2}\pi - \theta$ and from $\frac{1}{2}\pi - \theta$ to $\frac{1}{2}\pi + \theta$. The first has the simple value

$$-4\cos\theta\sin\left(\frac{1}{4}\pi - \frac{1}{2}\theta\right).$$

The second, consisting of two parts, is, before taking the limits into account,

$$\frac{\sin\theta}{\pi}\int\frac{\sin\psi_1\cos\lambda}{\sin\frac{1}{2}\lambda}\,d\lambda - \frac{2\cos\theta}{\pi}\int\psi_1\cos\frac{1}{2}\lambda\,d\lambda.$$

Integration by parts in the last term gives it the form

$$\frac{\sin\theta}{\pi}\int\frac{\sin\psi_1\cos\lambda}{\sin\frac{1}{2}\lambda}\,d\lambda + \frac{4\cos\theta}{\pi}\int\frac{\partial\psi_1}{\partial\lambda}\sin\frac{1}{2}\lambda\,d\lambda - \frac{4}{\pi}\psi_1\cos\theta\sin\frac{1}{2}\lambda.$$

Since the equation determining ψ_1 gives, at the two limits, respectively $\psi_1 = \pi$ and $\psi_1 = 0$, the term free of the integral sign contributes, on passing to the limits, a value opposite to that found above for the first integral. Thus only the first two terms remain. Substitution of the expressions obtained from the equation for ψ_1,

$$\sin\psi_1 = \frac{\sqrt{\sin^2\theta - \cos^2\lambda}}{\sin\theta\sin\lambda}, \qquad \frac{\partial\psi_1}{\partial\lambda} = -\frac{\cos\theta}{\sin\lambda}\frac{1}{\sqrt{\sin^2\theta - \cos^2\lambda}},$$

gives them the form

$$\frac{1}{\pi}\int\frac{\cos\lambda}{\sin\lambda\sin\frac{1}{2}\lambda}\sqrt{\sin^2\theta - \cos^2\lambda}\,d\lambda - \frac{4\cos^2\theta}{\pi}\int\frac{\sin\frac{1}{2}\lambda}{\sin\lambda}\frac{d\lambda}{\sqrt{\sin^2\theta - \cos^2\lambda}},$$

where the integrals are to be taken from $\lambda = \frac{1}{2}\pi - \theta$ to $\lambda = \frac{1}{2}\pi + \theta$. Introducing as new variable

$$z = \frac{\cos\lambda}{\sin\theta},$$

the limits for it become $+1$ and -1; changing the sign, they become -1 and $+1$. Substitution in the expression for ρ, while also taking account of the vanishing of $\varphi(\pi)$, gives

$$\rho = \frac{1}{2\sqrt{2}\pi^2}\int_{-1}^{+1}\left(\left(\frac{2 - z\sin\theta}{1 - z^2\sin^2\theta}\right)\cos^2\theta - z\sin\theta\right)\frac{dz}{\sqrt{(1-z^2)(1 - z\sin\theta)}}.$$

This result only appears to contain a component depending on elliptic integrals of the third kind. Since the limits are two consecutive values for which the radical vanishes, the expression, brought to the usual form, will contain only so-called complete elliptic integrals[9] and hence, by a beautiful theorem due to Legendre, can be reduced to the first and second kinds.

Finally, let us note how the case of the problem treated in the paper mentioned above, for a surface which differs only slightly from a spherical surface, can be reduced to the case of the sphere without a series expansion.

Let

$$r = 1 + \gamma z$$

[9] Fundamenta nova theoriae functionum ellipticarum, by C. G. J. Jacobi, p. 14. C. G. J. Jacobi's collected works, vol. I, p. 67. K.

be the equation of the surface, where γ is a small constant whose higher powers are to be neglected, and z denotes a function of θ and φ. Agree to denote by an added accent the value which a function of θ, φ takes when θ, φ are replaced by θ', φ'. Let ds' be the element of the surface belonging to θ', φ', and let p be the distance between the two points of the surface corresponding to the coordinate pairs θ, φ and θ', φ'. The problem requires that the equation

$$V = \int \rho' \frac{ds'}{p}$$

be satisfied for all values of θ, φ in the same range, the double integration extending over the whole surface, or from $\theta' = 0$, $\varphi' = 0$ to $\theta' = \pi$, $\varphi' = 2\pi$. If $d\sigma'$ is the element of the spherical surface corresponding to ds', cut off by the same radius vector as ds', and if q is the distance between the points on the spherical surface belonging to θ, φ and θ', φ', then the simplest geometrical considerations immediately give

$$ds' = (1 + 2\gamma z')d\sigma'$$

and

$$p = q\left(1 + \frac{1}{2}\gamma(z + z')\right).$$

Substitution of these expressions turns our equation into

$$V = \int \rho' \left(1 + \gamma\left(\frac{3}{2}z' - \frac{1}{2}z\right)\right)\frac{d\sigma'}{q},$$

or

$$V + \frac{1}{2}\gamma z \int \rho' \frac{d\sigma'}{q} = \int \rho' \left(1 + \frac{3}{2}\gamma z'\right)\frac{d\sigma'}{q}.$$

Since, by the penultimate equation, the integral multiplied by γ in the last equation differs from V only by a quantity of order γ, V may be put for it. One obtains

$$\left(1 + \frac{1}{2}\gamma z\right)V = \int \rho' \left(1 + \frac{3}{2}\gamma z'\right)\frac{d\sigma'}{q},$$

which reduces the problem to the case of the spherical surface. Thus one sees that the given potential value, multiplied by $1 + \frac{1}{2}\gamma z$, is to be regarded as applying to the spherical surface, and that the density determined on this assumption is then to be divided by $1 + \frac{3}{2}\gamma z$.

VIII.

On the Representability of Numbers by Three Squares.

By G. Lejeune Dirichlet.

Crelle, Journal für die reine und angewandte Mathematik, vol. 40, pp. 228–232.

On the Representability of Numbers by Three Squares.

The theory of decomposing the integers n which are not of one of the forms $4k$, $8k+7$ into three squares without a common divisor belongs to the more complicated parts of higher arithmetic, if one wishes to develop this theory completely, that is, not merely to prove representability, but also to determine the number of all possible decompositions. This number can be expressed either by means of the binary forms of determinant $-n$, as Gauss first showed[1], or also independently of the latter[2]. Since, however, there are cases in which only representability is to be assumed, as for example in the proof given by Cauchy[3], later simplified by Legendre, of Fermat's theorem on polygonal numbers, which rests solely on the fact that every number, apart from those excluded above, is the sum of three squares, a simple proof of representability seems not to be without interest. Such a proof is to be communicated in this paper.

For this purpose one first needs the known theorem that every positive ternary form of determinant -1, that is, every expression

$$ax^2 + by^2 + cz^2 + 2a'yz + 2b'xz + 2c'xy, \tag{1}$$

whose integral coefficients satisfy the equation

$$aa'^2 + bb'^2 + cc'^2 - abc - 2a'b'c' = -1 \tag{2}$$

and, moreover, satisfy the conditions that

$$a, \quad b, \quad c, \quad bc - a'^2, \quad ac - b'^2, \quad ab - c'^2$$

are positive (of which conditions, incidentally, the first and fourth imply the others), is equivalent to the form

$$x^2 + y^2 + z^2. \tag{3}$$

To prove this theorem it suffices to verify that, for determinant -1, the expression (3) is the only reduced positive form.

If the form (1) is to be reduced, then, by the theorem proved at the end of the preceding paper[4], abc must not be greater than 2. Hence, if one assumes $a \le b \le c$, it follows at once that

$$a = b = 1.$$

[1] Disquisitiones arithmeticae, art. 229.

[2] Crelle's Journal, vol. 21, p. 155. The editor of the Werke adds: p. 496 of vol. I of this edition of G. Lejeune Dirichlet's works. K.

[3] Exercices de Mathématiques par Cauchy, année 1826, p. 265. The editor of the Werke adds: Oeuvres complètes d'Augustin Cauchy, IIe Série, Tome VI, p. 320. K.

[4] Vol. II, p. 48 of this edition of G. Lejeune Dirichlet's works. The cited paper, "Über die Reduction der positiven quadratischen Formen mit drei unbestimmten ganzen Zahlen", is printed in vol. 40 of Crelle's Journal immediately before this paper. K.

On the other hand, by the definition of reduced forms, both $2c'$ and $2b'$, apart from sign, must not be greater than a, and $2a'$ must not be greater than b; thus one obtains

$$a' = b' = c' = 0,$$

and then, from (2),

$$c = 1.$$

This being assumed, the decomposability of the positive number a will have been shown as soon as one has found a positive ternary form (1) of determinant -1 whose first coefficient is a. For since such a form is equivalent to the form (3), the latter can be transformed into it, and one obtains

$$a = \alpha^2 + \alpha'^2 + \alpha''^2,$$

where α, α', α'' are three of the nine substitution coefficients and cannot have a common divisor, since every term of the determinant formed from these nine coefficients has either α, or α', or α'' as a factor, and this determinant is equal to unity.

Thus everything comes down to the following: regarding a as given, satisfy equation (2) by five integers b, c, a', b', c', with only the single remaining condition that $bc - a'^2$ must become positive. If one takes $b' = 1$, $c' = 0$, the equation becomes

$$b = a\Delta - 1,$$

where

$$\Delta = bc - a'^2$$

is set; and it remains only to show that the positive number Δ can be chosen so that $-\Delta$ is a quadratic residue of $b = a\Delta - 1$, since under this assumption c and a' can be so determined that $a'^2 - bc = -\Delta$.

We shall now show that the condition just stated can always be satisfied by an odd value of Δ such that, if a has the form $4k + 2$, then $b = a\Delta - 1$ is an odd prime number p, and if a is odd but not of the form $8k + 7$, then b is twice such an odd prime number p.

We begin with the second case. We have the equation

$$2p = a\Delta - 1,$$

where for the moment p is not yet assumed prime, but only odd. Put $\Delta = 8t + \varepsilon$, where ε is one of the numbers 1, 3, 5, 7. One sees at once that in each of the four cases presented by a with respect to the divisor 8, Δ admits two forms with respect to this divisor, or ε admits two values, if p, as required, is to be odd. For each of the eight cases thus obtained, apply the law of reciprocity to the left side of the equation following from $2p = a\Delta - 1$,

$$\left(\frac{p}{\Delta}\right) = \left(\frac{-2}{\Delta}\right),$$

where the Legendre symbol is used in the extended sense introduced by Jacobi; put for $\left(\frac{-2}{\Delta}\right)$ its known value, and then multiply by $\left(\frac{-1}{p}\right) = \pm 1$, where ± 1 will likewise be known according to the corresponding linear form of p. One obtains the following results:

$$
\begin{aligned}
a &= 8k+1, & &\begin{cases} \Delta = 8t+3, & p = 4s+1, \quad \left(\frac{-\Delta}{p}\right) = +1, \\ \Delta = 8t+7, & p = 4s+3, \quad \left(\frac{-\Delta}{p}\right) = -1, \end{cases} \\
a &= 8k+3, & &\begin{cases} \Delta = 8t+1, & p = 4s+1, \quad \left(\frac{-\Delta}{p}\right) = +1, \\ \Delta = 8t+5, & p = 4s+3, \quad \left(\frac{-\Delta}{p}\right) = +1, \end{cases} \\
a &= 8k+5, & &\begin{cases} \Delta = 8t+3, & p = 4s+3, \quad \left(\frac{-\Delta}{p}\right) = +1, \\ \Delta = 8t+7, & p = 4s+1, \quad \left(\frac{-\Delta}{p}\right) = -1, \end{cases} \\
a &= 8k+7, & &\begin{cases} \Delta = 8t+1, & p = 4s+3, \quad \left(\frac{-\Delta}{p}\right) = -1, \\ \Delta = 8t+5, & p = 4s+1, \quad \left(\frac{-\Delta}{p}\right) = -1. \end{cases}
\end{aligned}
$$

It follows that, if, as assumed, a is not of the form $8k+7$, the condition $\left(\frac{-\Delta}{p}\right) = 1$ can always be met. That in all eight cases p can also be made a prime number is clear from the expression

$$p = \frac{1}{2}(a\Delta - 1) = 4at + \frac{1}{2}(a\varepsilon - 1),$$

in which $\frac{1}{2}(a\varepsilon - 1)$ is, by the choice of ε, odd and also has no common divisor with a. This expression is therefore the general term of an arithmetic progression which necessarily contains prime numbers. Once one has a prime number p for which $\left(\frac{-\Delta}{p}\right) = 1$, $-\Delta$ is a quadratic residue of p and consequently also of $2p$, as was required.

In the case of an even a we at once put $a = 4k+2$, since one knows beforehand that for $a = 4k$ the condition cannot be satisfied. Since in the equation

$$p = a\Delta - 1$$

we assume Δ to be odd, p has the form $4s+1$, and we obtain

$$\left(\frac{-1}{\Delta}\right) = \left(\frac{p}{\Delta}\right) = \left(\frac{\Delta}{p}\right) = \left(\frac{-\Delta}{p}\right).$$

Thus Δ must be given the form $4t+1$ in order that $\left(\frac{-\Delta}{p}\right) = 1$. One obtains

$$p = 4at + a - 1,$$

and this expression can again become a prime number, for which $-\Delta$ is then a quadratic residue.

The application of the procedure just developed is not restricted to the case of determinant -1, and we add a second example for determinant -3.

If one again seeks, as above, the reduced forms for this determinant, then the condition

$$abc \leq 6$$

gives, under the assumption $a \leq b \leq c$, the value 1 for a, while b can be equal to 1 or to 2. It follows, since $2c'$ and $2b'$ must not be numerically greater than $a = 1$, that

$$b' = c' = 0,$$

and the equation by which the determinant is determined reduces to

$$a'^2 - bc = -3.$$

Now, since $2a'$ also must not be numerically greater than b, if $b = 1$ then $a' = 0$, and consequently $c = 3$. If, on the other hand, $b = 2$, then $2a' = 0$ or $2a' = \pm 2$. The first value does not satisfy the above equation, in which $bc = 4$. It remains only to set $a' = \pm 1$, whence $c = 2$. Neglecting the lower sign, which amounts only to changing the sign of z, one therefore obtains the two reduced forms

$$x^2 + y^2 + 3z^2, \qquad x^2 + 2y^2 + 2z^2 + 2yz, \tag{4}$$

so that every positive ternary form (1) of determinant -3, that is, one in which the coefficients satisfy the equation

$$aa'^2 + bb'^2 + cc'^2 - abc - 2a'b'c' = -3, \tag{5}$$

will be equivalent to one of these forms (4). If one now examines the residues modulo 8 which the expressions (4) leave when the elements x, y, z are given all combinations of even and odd values, excluding only the case of three even values simultaneously, since the elements are always to be assumed without a common divisor, one finds that the first of the forms (4) offers all possible residues modulo 8, whereas the second expression assumes neither of the forms $8k + 5$, $4k$ for any combination. If one further considers that two equivalent forms always represent the same numbers, then, for a form of determinant -3 which represents a number $8k + 5$ or $4k$ (which is in particular always the case when one of its first three coefficients, for instance a, is such a number), one can at once conclude that it is not equivalent to the second form, and consequently that it is equivalent to the first of the forms (4).

It remains to prove that the given number a is representable by the first of the forms (4). By the preceding method, and by again putting in (5)

$$b' = 1, \qquad c' = 0, \qquad \Delta = bc - a'^2,$$

we need only show that the positive number Δ can be so determined that $b = a\Delta - 3$ has the form $8k + 5$ or $4k$, and that $-\Delta$ is at the same time a quadratic residue of b. We restrict ourselves here to the numbers a which have no common divisor with the determinant, that is, which are not divisible by 3. Put

$$\Delta = 8\Delta',$$

where Δ' is to be odd and not divisible by 3. Then b will be relatively prime to Δ' and will have the form $8k + 5$, and only the other condition remains to be fulfilled. From the equation $b = 8a\Delta' - 3$ it follows at once that

$$\left(\frac{b}{\Delta'}\right) = \left(\frac{-3}{\Delta'}\right),$$

or, if one takes account of the congruence $b \equiv 1 \pmod 4$ and applies known theorems,

$$\left(\frac{b}{\Delta'}\right) = \left(\frac{\Delta'}{b}\right) = \left(\frac{-\Delta'}{b}\right) = \left(\frac{-3}{\Delta'}\right).$$

Multiplication by $\left(\frac{8}{b}\right) = \left(\frac{2}{b}\right) = -1$ gives from this

$$\left(\frac{-\Delta}{b}\right) = -\left(\frac{-3}{\Delta'}\right),$$

and one sees that the condition $\left(\frac{-\Delta}{b}\right) = 1$ is fulfilled if

$$\Delta' = 6t - 1$$

and consequently

$$b = 48at - 8a - 3$$

is set. Since this latter expression can evidently be made equal to a prime number, it is proved that every number not divisible by 3 can be expressed by the form

$$x^2 + y^2 + 3z^2$$

in such a way that x, y, z have no common divisor. Similar considerations can be applied to the numbers divisible by 3, for whose representability an easily obtained condition is required, as well as to the numbers expressible by the second of the forms (4).

Berlin, June 1850.

IX.

On a Problem Concerning Division.

By G. Lejeune Dirichlet.

Report on the Proceedings of the Royal Prussian Academy of Sciences, 1851, pp. 20–25.

On a Problem Concerning Division.

[Communicated in the meeting of the physical-mathematical class of the Academy of Sciences on 20 January 1851.[1]]

In a paper[2] belonging to the year 1849, it was incidentally observed that, in dividing an integer n by all divisors not larger than it, the case in which the remainder lies below half the divisor occurs more often than the opposite case, in which the remainder exceeds or equals half the divisor. It was also shown there that, as n increases, the ratio of the number of divisors for which the first case occurs to their total number n tends to the limit $2 - \log 4 = 0.61370\ldots$. It seems of some interest to generalize the investigation and to determine the number h of those divisors $1, 2, \ldots, p$, where $p \leq n$, for which the corresponding remainder has ratio to the divisor less than a given proper fraction α. If square brackets are used to denote the greatest integer contained in the enclosed value, so that $x - [x]$ is always zero or a positive proper fraction, then it is easy to see that the divisor s will or will not have the required property according as the difference

$$\left[\frac{n}{s}\right] - \left[\frac{n}{s} - \alpha\right]$$

is equal to the positive unit or to zero. Thus one has

$$h = \sum_1^p \left(\left[\frac{n}{s}\right] - \left[\frac{n}{s} - \alpha\right]\right) = \sum_1^p \left[\frac{n}{s}\right] - \sum_1^p \left[\frac{n}{s} - \alpha\right],$$

where, here and throughout what follows, the summation sign refers to s. In this form the expression for h is neither well suited to numerical computation, nor does it show how h changes for increasing values of n and p. A form serving this double purpose is obtained by the following transformation, which is also applicable in many other cases.

Let $y = f(x)$ be a function which continually decreases as the variable x increases from $x = \mu$ to $x = p$. The inverse function $x = F(y)$ will plainly have the same character and will likewise continually decrease, while the variable y increases from $y = f(p)$ to $y = f(\mu)$. Taking the constants μ and p to be integers, put, for abbreviation,

$$[f(\mu)] = \nu, \qquad [f(p)] = q,$$

and form the sequence

$$[f(\mu)],\ [f(\mu+1)],\ \ldots,\ [f(s)],\ \ldots,\ [f(p)],$$

[1] In the 1851 volume of the Academy reports the communication is introduced on p. 20 with the words: Herr Lejeune Dirichlet laid before the class the following note on a problem concerning the theory of division. In the printing on pp. 151–154 of volume 47 of Crelle's Journal (1854), Dirichlet altered the title and text somewhat. I have here used this later version almost throughout. K.

[2] On the determination of mean values in number theory. Vol. II, p. 49 of this edition of G. Lejeune Dirichlet's Works. The cited remark is found on p. 57. K.

in which each term is equal to or exceeds the following one. We shall now determine which terms of this sequence are equal to an arbitrary integer t lying between q and ν.

For this purpose first seek the completely determined index s of the term whose value is $\geq t$, while the following term is $< t$. Thus one has

$$[f(s)] \geq t, \qquad [f(s+1)] < t,$$

or, what is the same thing,

$$f(s) \geq t, \qquad f(s+1) < t,$$

and from the hypothesis made on the function $f(x)$ it follows that

$$s \leq F(t), \qquad s+1 > F(t),$$

hence

$$s = [F(t)]\,.$$

Applying this result to t and $t+1$, one sees that the value t belongs only to those terms whose index s satisfies the double condition

$$s > [F(t+1)]\,, \qquad s \leq [F(t)]\,.$$

Because the beginning and end of the sequence are prescribed, this result is modified for $t = \nu$ by requiring the first condition to be $s \geq \mu$, and for $t = q$ by replacing the second condition by $s \leq p$.

Taking this result into account, it is now easy to transform the sum

$$\sum_{\mu+1}^{p} [f(s)]\varphi(s),$$

where $\varphi(s)$ denotes an entirely arbitrary function: first combine all terms for which $[f(s)]$ has one and the same value, and then add all the partial sums so obtained. Put

$$\sum_{1}^{s} \varphi(s) = \Psi(s).$$

Then the partial sum in which $[f(s)]$ has the value t is

$$\begin{aligned}
&t\big(\Psi[F(t)] - \Psi[F(t+1)]\big), && \text{if } q < t < \nu,\\
&\nu\big(\Psi[F(\nu)] - \Psi(\mu)\big), && \text{if } t = \nu,\\
&q\big(\Psi(p) - \Psi[F(q+1)]\big), && \text{if } t = q,
\end{aligned}$$

and therefore

$$\sum_{\mu+1}^{p} [f(s)]\varphi(s) = q\Psi(p) - \nu\Psi(\mu) + \sum_{q+1}^{\nu} \Psi[F(s)].$$

Now separate, in each of the two sums contained in the expression for h given above, the first μ terms, and apply the formula just found to the remaining terms. This gives

$$\sum_{\mu+1}^{p} \left[\frac{n}{s}\right] = \sum_{q+1}^{\nu} \left[\frac{n}{s}\right] + pq - \mu\nu,$$

where $\left[\frac{n}{\mu}\right] = \nu$ and $\left[\frac{n}{p}\right] = q$, and

$$\sum_{\mu+1}^{p} \left[\frac{n}{s} - \alpha\right] = \sum_{q'+1}^{\nu'} \left[\frac{n}{s+\alpha}\right] + pq' - \mu\nu',$$

where $\left[\frac{n}{\mu}-\alpha\right]=\nu'$ and $\left[\frac{n}{p}-\alpha\right]=q'$.

For abbreviation denote $\nu-\nu'$ by δ and $q-q'$ by ε, so that δ and ε can have only the values 0 or 1. Put the last sum in the form

$$\sum_{q'+1}^{\nu'}\left[\frac{n}{s+\alpha}\right]=\sum_{q+1}^{\nu}\left[\frac{n}{s+\alpha}\right]+\varepsilon\left[\frac{n}{q+\alpha}\right]-\delta\left[\frac{n}{\nu+\alpha}\right],$$

and substitute. One obtains

$$\begin{aligned}h=&\sum_{1}^{\mu}\left(\left[\frac{n}{s}\right]-\left[\frac{n}{s}-\alpha\right]\right)+\sum_{q+1}^{\nu}\left(\left[\frac{n}{s}\right]-\left[\frac{n}{s+\alpha}\right]\right)\\&+\left(p-\left[\frac{n}{q+\alpha}\right]\right)\varepsilon-\left(\mu-\left[\frac{n}{\nu+\alpha}\right]\right)\delta.\end{aligned}$$

Although this transformation is valid for all values of p, it is advantageous only when p is greater than $\sqrt{n}$. Under this hypothesis it is most advantageous when, as will be done below, one chooses for the number μ, which has so far been arbitrary, one of the integers adjacent to $\sqrt{n}$. As is easy to see, the number of divisions needed for the exact determination of h is then approximately

$$2\sqrt{n}-\frac{n}{p},$$

whereas the original expression required p divisions.

We shall now, under the hypothesis that p is of higher order than $\sqrt{n}$, that is, that $\frac{p}{\sqrt{n}}$ grows beyond every bound with n, seek to determine the limiting value of the ratio $\frac{h}{p}$ of the number of divisors having the required property to their total number p. In this investigation one may neglect, in the expression for h, all terms whose order is lower than that of p. Dropping the first, whose order does not exceed $\sqrt{n}$, and also the fourth, which can contain only a bounded number of units, one obtains

$$h=\sum_{q+1}^{\nu}\left(\left[\frac{n}{s}\right]-\left[\frac{n}{s+\alpha}\right]\right)+\left(p-\left[\frac{n}{q+\alpha}\right]\right)\varepsilon,$$

or also, if one omits the square brackets, which plainly causes only a change whose order does not exceed $\sqrt{n}$,

$$h=n\sum_{q+1}^{\nu}\left(\frac{1}{s}-\frac{1}{s+\alpha}\right)+\left(p-\frac{n}{q+\alpha}\right)\varepsilon.$$

If the upper limit ν is changed to ∞, the sum receives the increment

$$\frac{1}{\nu+1}-\frac{1}{\nu+1+\alpha}+\frac{1}{\nu+2}-\frac{1}{\nu+2+\alpha}+\cdots<\frac{1}{\nu+1},$$

whose value, multiplied by n, likewise does not exceed the order $\sqrt{n}$. Thus one obtains

$$\lim\frac{h}{p}=\frac{n}{p}\sum_{q+1}^{\infty}\left(\frac{1}{s}-\frac{1}{s+\alpha}\right)+\left(p-\frac{n}{q+\alpha}\right)\frac{\varepsilon}{p}.$$

It remains to distinguish the case where the quotient $\frac{n}{p}$, which by hypothesis is ≥ 1, grows beyond every bound, from the case where this quotient remains finite.

In the first case the second term approaches zero, while the ratio of the sum contained in the first term to $\frac{\alpha}{q}$ tends to unity. Thus the limit of $\frac{h}{p}$ coincides with that of $\frac{n}{pq}\alpha$, that is, with α.

In the second case, where $\frac{n}{p}$, and hence also $q = \left[\frac{n}{p}\right]$, remains finite, it is expedient to bring the limiting value of $\frac{h}{p}$ given directly by the last equation into another form. Instead of the sum one introduces the difference of two others, taken from $s = 1$ to respectively $s = \infty$ and $s = q$, and then expresses the first by an integral. Our equation then becomes

$$\lim \frac{h}{p} = \frac{n}{p} \int_0^1 \frac{1 - \varphi^\alpha}{1 - \varphi}\, d\varphi - \frac{n}{p} \sum_1^q \left(\frac{1}{s} - \frac{1}{s + \alpha}\right) + \left(1 - \frac{n}{p}\frac{1}{q + \alpha}\right)\varepsilon,$$

where the integral, which for every rational value of α can be represented by logarithms and circular functions, is a known transcendental function studied many times. If one sets in particular $p = n$, then $q = 1$, $q' = 0$, $\varepsilon = 1$, and the limiting expression passes into

$$\lim \frac{h}{n} = \int_0^1 \frac{1 - \varphi^\alpha}{1 - \varphi}\, d\varphi.$$

With the help of the table of these transcendents given by Gauss in the paper *Disquisitiones generales circa seriem infinitam* …[3], one can easily determine the value of α corresponding to a given value of the integral. For example, one finds that, if for half the divisors $1, 2, \ldots, n$ the ratio of the remainder to the divisor is to lie below α, then $\alpha = 0.384686\ldots$ is required.

[3] Gauss' Works, vol. III, p. 161. K.

X.

On the Composition of Binary Forms of the Second Degree.

P. G. Lejeune Dirichlet.

Commentary, Berlin, Academic Press, 1851.

On the Composition of Binary Forms of the Second Degree[1].

Several years ago, when the problem had arisen of transferring to the theory of complex integers the question of the number of classes of forms of the second degree corresponding to a given determinant, I had to set out the elements of the theory of forms from the beginning, since these had previously been developed only in the case of real integers. With the aid of certain considerations not previously used in this theory, and equally suited to real and complex integers, I succeeded in completing that exposition of the elements in a few pages[2]. There I restricted the investigation to those properties which concern equivalence of forms, transformation, and representation of numbers by forms, since these alone were required for the question to which that paper was devoted. I did not then treat the composition of forms, an argument that the illustrious Gauss, in the fifth section of the *Disquisitiones arithmeticae*, is known to have treated with the greatest generality, but by computations so lengthy that very few have been able to grasp the nature of composition, all the more because the great geometer, as he himself noted, arranged the demonstrations of the more difficult theorems synthetically in the interest of brevity, suppressing the analysis by which they had been discovered. I therefore believe I may trust that a new and wholly elementary exposition of this topic will not be unwelcome to students of analysis.

I.

Before we approach the composition of forms, a few facts, either known or easily deduced from known ones, must be stated first.

Given values $\zeta, \zeta', \zeta'', \ldots$ satisfying the congruence

$$u^2 \equiv D$$

with respect respectively to the moduli $m, m', m'', \ldots$, we shall call them concordant with one another if a root Z of the same congruence, referred to the modulus $mm'm'' \ldots$, can be found such that

$$\zeta \equiv Z \pmod{m}, \qquad \zeta' \equiv Z \pmod{m'}, \qquad \zeta'' \equiv Z \pmod{m''}, \ldots.$$

It will suffice to consider the case in which the moduli $m, m', m'', \ldots$ are odd and prime to D itself.

It is easy to see that the proposed congruences cannot be satisfied unless, with respect to each prime number dividing two or more of $m, m', m'', \ldots$, the corresponding values $\zeta, \zeta', \zeta'', \ldots$ are

[1] In the printing on pp. 155–160 of volume 47 of Crelle's Journal (1854), Dirichlet altered the text in several places. I have here used this later version almost throughout. K.

[2] *Recherches sur les formes quadratiques à coefficients et à indéterminées complexes*, in Crelle's Journal, vol. XXIV. Vol. I, p. 533 of this edition of G. Lejeune Dirichlet's Works. K.

congruent to one another. If this condition holds, then from the given values $\zeta, \zeta', \zeta'', \ldots$ one knows the residues $\pi, \pi', \pi'', \ldots$ of Z with respect to the distinct prime numbers $p, p', p'', \ldots$ dividing the product $mm'm'' \ldots$. Since the residues $\pi, \pi', \pi'', \ldots$ are plainly roots of our congruence with respect respectively to the moduli $p, p', p'', \ldots$, it follows from the elementary theory of congruences that there exists a root, and in fact a unique root, Z satisfying the congruence modulo $mm'm'' \ldots$ and congruent to $\pi, \pi', \pi'', \ldots$ modulo $p, p', p'', \ldots$ respectively. At the same time it is immediate that

$$Z \equiv \zeta \pmod{m}, \qquad Z \equiv \zeta' \pmod{m'}, \qquad Z \equiv \zeta'' \pmod{m''}, \ldots.$$

Since the constant term D occurring in our congruence is to retain the same value throughout what follows, we shall, for brevity, denote a given root ζ corresponding to the modulus m by the notation (m, ζ). With this notation introduced, the root Z to be deduced in the indicated way from the mutually concordant given roots $\zeta, \zeta', \zeta'', \ldots$, and which we shall say is composed from them, may conveniently be denoted as follows:

$$(m, \zeta)(m', \zeta')(m'', \zeta'') \cdots = (mm'm'' \ldots, Z).$$

We further observe that the roots $\zeta, \zeta', \zeta'', \ldots$ are always mutually concordant if $m, m', m'', \ldots$ are relatively prime to one another; and this remains true in the case that we excluded, in which $m, m', m'', \ldots$ are not prime to $2D$. Indeed, then Z is already completely determined modulo $mm'm'' \ldots$ by the congruences

$$Z \equiv \zeta \pmod{m}, \qquad Z \equiv \zeta' \pmod{m'}, \qquad Z \equiv \zeta'' \pmod{m''}, \ldots,$$

and at the same time one has

$$Z^2 \equiv D \pmod{mm'm'' \ldots}.$$

II.

For the forms of the second degree considered here,

$$ax^2 + 2bxy + cy^2,$$

whose determinant is $D = b^2 - ac$, we shall suppose the coefficients a, b, c to be free of a common divisor, since the remaining cases are easily reduced to this one. It is known that forms satisfying the stated condition constitute two orders, according as at least one of the outer coefficients a, c is odd or both are even. For brevity we shall exclude the latter case here, since the arguments applicable to the former apply, with the necessary changes, in the other as well.

If, in the given form, definite relatively prime values are assigned to the indeterminates x, y (which will always be understood), so that

$$ax^2 + 2bxy + cy^2 = m,$$

we shall say that the number m (which is always to be supposed prime to $2D$) is represented by the form. Taking integers ξ, η such that $x\eta - y\xi = 1$, it is known and easily proved that the expression

$$\zeta = (ax + by)\xi + (bx + cy)\eta$$

is a root of the congruence $u^2 \equiv D \pmod{m}$, and that the same root is always obtained in this way, however ξ and η may vary. The root (m, ζ) to which we shall say the given representation belongs can be defined in another way, more suited to our purpose. In the equation by which we have displayed ζ, multiply by y and put $x\eta - 1$ in place of the product $y\xi$; then one sees that

$$ax + by \equiv -y\zeta \pmod{m}. \tag{1}$$

This congruence fully determines ζ, provided that y is prime to the modulus m. If, however, y has with the modulus a greatest common divisor δ greater than unity, which plainly also divides a, our congruence gives only

$$\frac{ax+by}{\delta} \equiv -\frac{y}{\delta}\zeta \pmod{\frac{m}{\delta}},$$

so that the residues of ζ cannot be known in this way with respect to those prime divisors of δ, if there are any, which do not divide $\frac{m}{\delta}$. This inconvenience is easily remedied if one observes that the equations above imply

$$\zeta \equiv bx\eta, \qquad x\eta \equiv 1 \pmod{\delta},$$

and hence

$$\zeta \equiv b \pmod{\delta}. \tag{2}$$

Together with the preceding formula, this now fully determines the residues of ζ with respect to the individual divisors of m. We observe that, if ε is a common divisor of x and m, one similarly obtains

$$\zeta \equiv -b \pmod{\varepsilon}. \tag{3}$$

We add the following facts; although they are sufficiently well known, it is useful to put them in view here.

1°. If two forms are equivalent, properly (as will always be understood), and the former passes into the latter by the substitution

$$x = \alpha x' + \beta y', \qquad y = \gamma x' + \delta y',$$

where $\alpha\delta - \beta\gamma = 1$, then clearly a number m representable by one of them can also be represented by the other. It is also easy to prove that the two representations of the same number, connected by the preceding equations, belong to the same root (m, ζ). Hence the representations belonging to a given expression (m, ζ) must be referred to one whole class, and that class is unique.

2°. Conversely, one can prove that if the same number m is represented by two forms of the same determinant, and the two representations belong to the same root (m, ζ), then the forms are equivalent.

3°. Finally, it is clear that, given any root (m, ζ), there exists a class by whose forms m can be represented in such a way that the root corresponding to these representations is (m, ζ). Indeed m is plainly represented by the form

$$\left(m, \zeta, \frac{\zeta^2 - D}{m}\right),$$

whose coefficients are free of a common divisor and in which m is odd, by putting

$$x = 1, \qquad y = 0;$$

it is evident that this representation belongs to (m, ζ).

III.

With these preliminaries completed, let us approach the proposed question. Let two forms φ and φ' of the same determinant D be given, and let m and m' be any two odd integers prime to D, representable respectively by φ and φ' in such a way that the roots (m, ζ) and (m', ζ') to which these representations belong are mutually concordant. Under these hypotheses, the whole matter turns on proving that the representations of mm' belonging to the expression

$(m,\zeta)(m',\zeta')$ will always be effected by the same form, or rather are to be referred to the same class, however m and m' may vary. This is the fundamental theorem in this theory.

Suppose that the given forms (a,b,c) and (a',b',c') have been so prepared that the expressions (a,b) and (a',b') are mutually concordant. This is achieved, for example, by transforming one of the forms into an equivalent one whose first coefficient has no common divisor with the first coefficient of the other. But it must be carefully noted that the analysis developed below requires nothing except that (a,b) and (a',b') be mutually concordant; it is not necessary that a and a' be relatively prime to one another, nor that they be prime to $2D$. Denote by (aa',B) the expression arising from the composition of (a,b) and (a',b'), so that

$$B \equiv b \pmod{a}, \qquad B \equiv b' \pmod{a'}, \qquad D = B^2 - aa'C,$$

where C is an integer. It is then clear that the forms can be changed into the following equivalent forms:

$$ax^2 + 2Bxy + a'Cy^2 = \varphi, \qquad a'x'^2 + 2Bx'y' + aCy'^2 = \varphi'.$$

First multiplying these respectively by a and a', and then multiplying the resulting equations together, one obtains

$$(ax+By)^2 - Dy^2 = a\varphi, \qquad (a'x'+By')^2 - Dy'^2 = a'\varphi',$$

$$((ax+By)(a'x'+By') + Dyy')^2 - D((ax+By)y' + (a'x'+By')y)^2 = aa'\varphi\varphi'.$$

Now, observing that $D = B^2 - aa'C$, one has

$$(ax+By)(a'x'+By') + Dyy' = aa'X + BY, \tag{4}$$

where

$$X = xx' - Cyy', \qquad Y = (ax+By)y' + (a'x'+By')y = axy' + a'x'y + 2Byy'. \tag{5}$$

The last equation, divided by aa', assumes the form

$$aa'X^2 + 2BXY + CY^2 = \varphi\varphi' = \psi. \tag{6}$$

After we have displayed indefinitely the product of the forms φ and φ' by the form ψ of the same determinant D, suppose that both x, y and x', y' are assigned definite relatively prime values for which $\varphi = m$ and $\varphi' = m'$, and which satisfy the conditions indicated above. Then it remains to prove: 1° that the representation of mm' by the form ψ, which we see is supplied by equations (5) and (6), will be proper, that is, that X and Y will be free of a common divisor; and 2° that the root to which this representation belongs, say (mm', Z), will indeed be composed from the roots (m,ζ) and (m',ζ') to which the representations of m and m' are assumed to belong.

1°. To show that X and Y have no common divisor, let us start from the supposition that a prime number p divides both, and see what follows. Since p must divide one of m, m', let us suppose, as we may, that m is a multiple of p. I then say that m' too may be supposed divisible by p. Since X and Y are divisible by p, it is clear, multiplying the first of equations (5) by $-ay'$ and the second by x' and adding, that an integer divisible by p is obtained:

$$(a'x'^2 + 2Bx'y' + aCy'^2)y = m'y.$$

If one were now to suppose that m' is not divisible by p, then y, and consequently also a, would be multiples of p. But then, from

$$xx' - Cyy' = X \equiv 0 \pmod{p},$$

and from the fact that x is prime to y, it follows that $x' \equiv 0$, whence finally

$$m' = a'x'^2 + 2Bx'y' + aCy'^2$$

is seen to be a multiple of p. Since p divides both m and m', by equation (1) we have

$$ax + By \equiv -y\zeta, \qquad a'x' + By' \equiv -y'\zeta' \pmod{p}.$$

Substitution of these formulas into the second of equations (5) gives the congruence

$$(\zeta + \zeta')yy' \equiv 0 \pmod{p}.$$

If $\zeta + \zeta' \equiv 0$, then $\zeta' \equiv -\zeta$, contrary to the hypothesis that the roots ζ and ζ' are concordant, namely that $\zeta' \equiv \zeta$. It remains to see what follows from either of the suppositions $y \equiv 0$ and $y' \equiv 0$, which are entirely similar. If y, and therefore also x' by the congruence $xx' - Cyy' \equiv 0$, were divisible by p, then by equations (2) and (3) we would have

$$\zeta \equiv B, \qquad \zeta' \equiv -B \pmod{p},$$

and just as above

$$\zeta + \zeta' \equiv 0 \pmod{p}.$$

It is therefore established that X and Y are relatively prime.

2°. We now prove that the root to which the representation of mm' belongs, and which we have denoted by (mm', Z), is indeed composed from (m, ζ) and (m', ζ'). By symmetry it suffices to show that $Z \equiv \zeta$ with respect to any prime divisor p of m. Since by equation (1)

$$ax + By \equiv -\zeta y \pmod{p},$$

equation (4) and the second of equations (5) readily give the congruences

$$(-\zeta(a'x' + By') + Dy')y \equiv aa'X + BY, \qquad (-\zeta y' + a'x' + By')y \equiv Y \pmod{p}.$$

Multiplying the latter by ζ and adding it to the former, and taking account of the formula $\zeta^2 \equiv D \pmod{m}$, one obtains

$$aa'X + BY \equiv -Y\zeta \pmod{p}.$$

Comparing this congruence with

$$aa'X + BY \equiv -YZ \pmod{p},$$

one sees that

$$Z \equiv \zeta \pmod{p},$$

provided p does not divide Y. It remains to consider the case in which Y is divisible by p. In this case, by equation (2), $Z \equiv B \pmod{p}$. If now y is also a multiple of p, then similarly $\zeta \equiv B \pmod{p}$, and hence

$$Z \equiv \zeta \pmod{p}.$$

If, however, y is not divisible by p, the congruence found above gives

$$a'x' + By' - \zeta y' \equiv 0 \pmod{p},$$

from which it is easily deduced that p is a factor of the product $a'm'$. For

$$a'm' = (a'x' + By')^2 - Dy'^2 = (a'x' + By')^2 - \zeta^2 y'^2 \equiv 0 \pmod{p}.$$

Two cases must now be distinguished. First suppose that a' is not divisible by p. Since in this case m' is divisible by p, one has

$$a'x' + By' + \zeta' y' \equiv 0 \pmod{p}.$$

Comparing this formula with the preceding one gives

$$(\zeta + \zeta')y' \equiv 0 \pmod{p},$$

which leads to an absurd conclusion. For since $\zeta + \zeta'$ cannot be a multiple of p, it would follow that $y' \equiv 0 \pmod{p}$, and hence, from the equation

$$m' = a'x'^2 + 2Bx'y' + aCy'^2,$$

that $a' \equiv 0 \pmod{p}$, contrary to the hypothesis. Therefore this case cannot occur. In the remaining case, where a' is divisible by p, the congruence

$$a'x' + By' - \zeta y' \equiv 0 \pmod{p}$$

gives

$$(B - \zeta)y' \equiv 0 \pmod{p}.$$

If p does not divide y', one has

$$\zeta \equiv B \pmod{p},$$

which agrees with the congruence $Z \equiv B \pmod{p}$. If, on the other hand, p divides y', and consequently also m', then by equation (2) one has

$$\zeta' \equiv B \pmod{p},$$

and, as above, one concludes that $\zeta \equiv B \pmod{p}$, since the roots ζ and ζ' are concordant.